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NEURAL OPTICAL FLOW

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Abstract

Neural optical flow (NOF) offers improved accuracy and robustness over existing OF methods for particle image velocimetry (PIV). Unlike other OF techniques, which rely on discrete displacement fields, NOF parameterizes the physical velocity field using a continuous neural-implicit representation. This formulation enables efficient data assimilation and ensures consistent regularization across views for stereo PIV. The neural-implicit architecture provides significant data compression and supports a space–time formulation, facilitating the analysis of both steady and unsteady flows. NOF incorporates a differentiable, nonlinear image-warping operator that relates particle motion to intensity changes between frames. Discrepancies between the advected intensity field and observed images form the data loss, while soft constraints, such as Navier–Stokes residuals, enhance accuracy and enable direct pressure inference from PIV images. Additionally, mass continuity can be imposed as a hard constraint for both 2D and 3D flows. Implicit regularization is achieved by tailoring the network's expressivity to match a target flow's spectral characteristics. Results from synthetic planar and stereo PIV datasets, as well as experimental planar data, demonstrate NOF's effectiveness compared to state-of-the-art wavelet-based OF and CC methods. Additionally, we highlight its potential broader applicability to techniques like background-oriented schlieren, molecular tagging velocimetry, and other advanced measurement systems.

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Chapter 1

Introduction

1.1 Particle Image Velocimetry

Particle image velocimetry (PIV) analyses have been significantly improved by advanced processing techniques, enhancing the accuracy and resolution of velocity measurements [3]. PIV is a non-intrusive technique commonly used to measure 2D–3D velocity fields from images of a particle-laden flow exposed to a pulsed laser sheet or slab [4]. First, a flow of interest is seeded with tracer particles. Then, a camera is set to record at at a high frame rate while a pulsed laser sheet or slab is used to illuminate the particles for a brief moment in each frame, such that the particles appear frozen in each image. Finally, sequential images are processed using a computer vision algorithm, cross-correlation (CC) algorithm, or similar to determine the particle displacement field and this field is converted to a velocity field, typically assuming a constant velocity between frames. Unlike intrusive and local velocity measurements from physical probes, the spatially resolved and global velocity fields obtained through PIV can reveal key flow features such as vortices, separation bubbles, and organized packets of coherent structures [5]. Furthermore, PIV can be used to evaluate the performance of engineering devices and validate computational models inferring other fields like pressure, vorticity, strain rate, and Reynolds stresses, which may be derived from velocity fields, in some circumstances [6, 7]. Despite the success of current methods, however, the need for more accurate measurements and greater resolution continues to drive the development of improved PIV techniques.

1.2 Background Oriented Schlieren

Background Oriented Schlieren (BOS) is an optical method used to measure the density field of a transparent medium. BOS relies on schlieren, which are local inhomogeneities within the optical properties of a transparent substance. Most people are familiar with the effects of schlieren, in phenomena such as the shimmery, mirage-like effect above hot pavement, or in the way subsurface regions of an object partially submerged in water often look “disconnected” from the exposed regions. These effects are due to local changes in the speed of light through a medium, causing refraction and thus a “bending” effect on the light rays. Traditionally, changes in local speed of light are rephrased in terms of a refractive index,

$$n = \frac{c_0}{c}, \tag{1.1}$$

where c is the local speed of light through a transparent medium, and $c_0 \approx 3 \times 10^8$ is the speed of light in a vacuum. Within gasses, where n is nearly unity, the refractive index can be approximated with the Gladstone-Dale equation,

$$n = 1 + G\rho, \tag{1.2}$$

where G is the Gladstone-Dale coefficient and ρ is the density [8].

Refraction at a discrete interface, such as an air/water boundary, can be described with Snell’s Law,

$$n_1 \sin(\varepsilon_1) = n_2 \sin(\varepsilon_2)\,, \tag{1.3}$$

where ε_1 and ε_2 are the angles of incidence and refraction, respectively, as illustrated in Fig. 1.1. As a ray of light passes from one medium to another, the angle between the ray and the normal of

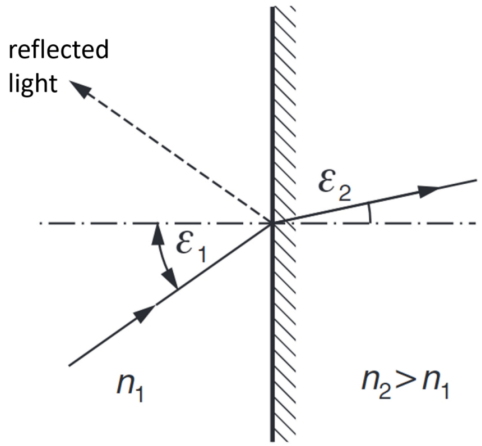

Figure 1.1: Visual example of Snell's Law, showcasing a light ray being partially reflected and partially refracted.

the interface will decrease with an increase in refractive index and vice versa. This causes the ray to bend, which is what gives rise to the distortion as one looks through water or glass. This principle can be generalized to media with continuously varying refractive indices, such as the previously mentioned "shimmer" distortion above hot pavement, where a temperature gradient in the air gives rise to a correlated gradient in density and thus refractive index. A continuous refractive index field can be thought of as infinitely many discrete transitions and a mathematical formulation can be derived from Snell's Law in the form of a path integral; for more details, see Settles [9].

BOS attempts to retrieve the density field of a fluid by measuring the optical effect that refractive index gradients have upon images taken through the transparent fluid of interest. A camera focuses through a working fluid onto an illuminated background plane featuring some pattern. A "reference" image is taken while the flow is quiescent and no density/refractive index gradients are present. Once the flow is activated, density gradients cause light to be refracted as it passes through the working fluid, resulting in a warped image of the background plate. In conjunction with the reference image taken while the flow was still, this warped image can be used to find deflections in the light rays as they pass through the fluid by use of a deflection sensing algorithm. With the continuous model of refraction, these deflections can be used to solve for the density field of the fluid projected onto the camera sensor, or a fully realized three-dimensional field, in the case of a tomographic setup. We only explore the deflection sensing aspect of BOS within this thesis, rather than the entire process.

1.3 Deflection Sensing Algorithms

Both PIV and BOS rely upon deflection sensing within their methodologies. Cross-correlation is the standard field inversion method for extracting velocity fields from pairs of particle images in PIV, and is less commonly used in BOS. In this approach, the first image in a pair is divided into smaller subsections called interrogation windows. For each interrogation window, a local search of nearby regions in the second image is performed to find the maximum correlation to the interrogation window in the first image, i.e. that looks the most like the interrogation window in the first image. The pixel displacement between the two regions is used to find the average

displacement of particles/intensity patterns inside the window [3]. A single velocity vector is generated for each interrogation window, and these windows are typically overlapped by 50–75% to produce a denser vector field. Multi-resolution techniques are typically used to improve the performance of CC by first using large interrogation windows to estimate a coarse velocity field, then estimating a finer displacement field with smaller windows, using the coarse estimate as a foundation, and then iterating this process to a final, optimal resolution. Hu et al. [10] found that optimal results are achieved when the smallest interrogation window is large enough to contain multiple particles but small enough to minimize excessive smoothing, typically ranging in size from 16×16 to 32×32 pixels. Regardless, even under optimal conditions, traditional CC can only yield fields with a vector spacing of about five pixels, causing velocity and gradient fields to appear smeared and low-pass filtered [10, 11, 12]. Despite significant efforts made to improve the resolution of CC-based measurements [13, 14, 15, 16, 11, 17], modern implementations still produce relatively coarse deflection fields. CC remains the most widely used technique for PIV analysis due to its computational speed, simplicity, and reliable performance, despite its poorer measurement accuracy.

Optical flow (OF) offers an alternative approach to deflection sensing for PIV/BOS by computing a displacement field that warps an image taken at time t to match the image from time $t + \Delta t$. OF algorithms generally generate a dense velocity field, with one vector per pixel,[1] making them particularly effective at resolving flows with broadband spectral content and sharp gradients. However, due to linearization approximations, classical OF methods are limited to handling small displacements [18], usually requiring a multi-resolution scheme to accommodate larger displacements [19, 20]. These schemes often struggle to resolve flows containing high spectral content and large gradients [21]. Hybrid approaches that integrate OF with CC have been proposed to address these issues, but they introduce additional parameters to tune and increase computational complexity [22, 19]. Since OF is inherently ill-posed, meaning there is not enough information within the images to fully solve for the displacements, regularization is required to obtain a stable solution; in most cases, the regularization parameters are heuristically tuned for each flow scenario [23]. Despite its advantages in accuracy and resolution, the adoption of OF for PIV has been limited due to its higher computational cost compared with CC, and its sensitivity to spatiotemporal variations in laser output and out-of-plane motion, both of which amplify errors. BOS has seen more widespread adoption of OF, as BOS deflection fields are often below a single pixel, which reduces the efficacy of CC. Furthermore, one is free to choose a background pattern more suited to OF, unlike in PIV, where imaging is always done on discrete points of localized intensity.

Wavelet-based optical flow (WOF) is one of the current state of the arts in OF for fluid dynamics research. Introduced by Wu et al. [24] and further developed by Dérian et al. [25], WOF attempts to take advantage of the multi-scale nature of a wavelet basis [26] to resolve turbulent flows with significant scale separation by representing the displacement field within said wavelet basis. Wavelet coefficients corresponding to coarse velocity features are trained first, followed by successive estimation of finer velocity features in a coarse-to-fine cascade. Regularization is applied either by truncating the transform, completely removing high frequency and spurious content alike, or adding a penalty term that promotes smoothness and stability. An estimation of the velocity field is used to warp the first image into the second, and the wavelet coefficients are optimized

[1] Although the number and arrangement of displacement vectors can vary, it is standard to estimate one vector per pixel.

via gradient descent, with a loss based on the discrepancy between the actual second image and the warped image; spatial gradients in the penalty term are computed directly in the wavelet domain [25, 27]. WOF consistently outperforms CC, as demonstrated by Schmidt and Sutton [12, 28] who benchmarked WOF against state-of-the-art CC methods across various flow scenarios and a range of particles-per-pixel (ppp) values. These tests demonstrated WOF's superior accuracy, especially at high wavenumbers and in capturing gradients, leading to more accurate vorticity fields. More recently, Jassal and Schmidt [29] and Nicolas et al. [30] highlighted WOF's advantages in recovering near-wall velocity content.

1.4 Machine Learning Based Field Inversion

The suitability of machine learning (ML) methods for image processing and its success in other fields has driven the rapid adoption of ML in fluid dynamics applications. These image processing strengths make ML a natural fit for analyzing optical data from fluid diagnostics. One of the earliest breakthroughs in ML for fluid applications involved convolutional neural networks (CNNs), which excel at processing image data by taking advantage of the inherent spatial information in images [31, 32, 33, 34, 35]. Building on this foundation, CNNs have been applied to PIV processing. Rabault et al. [36] and Lee et al. [37] developed CNN models that replaced the peak searching step of CC, but still produce a sparse velocity field with one vector per image window, e.g., 32×32 pixels in size. Cai et al. [38, 39] augmented this approach to achieve dense motion estimation, generating one vector per pixel with accuracy comparable to Horn–Schunck OF [39]. However, CNN-based algorithms require extensive pre-training on large, labeled datasets, typically comprising synthetic PIV images produced via direct numerical simulations (DNSs). This dependence on specific training data introduces the risk of overfitting and limits the generalizability of the trained models to unseen flow scenarios.

Conversely to previous methods that attempt to replace the decision making algorithm with an ML model, neural-implicit fields are an emerging paradigm in computer vision and scientific ML that replace a discrete basis, like pixels, with a neural network, allowing for unsupervised learning [40, 41, 42, 43]. These networks, termed coordinate neural networks, take spatial and/or temporal coordinates as inputs and output a quantity of interest at those coordinates. This neural basis provides significant data compression, particularly for spatio-temporally resolved fields, as compared to discrete bases [41]. Exact partial derivatives of the networks can be efficiently computed via automatic differentiation (AD), making these representations ideal for gradient-based optimization and data assimilation [44, 45], where the inferred fields are known to satisfy differential equations. Unlike CNNs and other supervised models, coordinate neural networks do not replace algorithmic aspects of computer vision methods[2], and thus can circumvent the need for labeled datasets, overcoming a key limitation of most ML methods used for PIV processing. Their versatility has led to applications in neural radiance fields for novel view synthesis [42, 43], tomographic imaging [46, 47, 48], and data assimilation [49, 50, 51]

Recently, Qi et al. [52] introduced a neural-implicit OF algorithm for PIV, demonstrating its ability to handle large displacements and enforce divergence free or irrotaional constraints through a Helmholtz decomposition. However, their method relies on "HDNet," which is essentially a

[2]Excluding differentiation, which is facilitated by AD with coordinate neural networks.

differentiable Poisson solver trained on extensive synthetic datasets. This component reintroduces challenges typical of supervised learning and may limit generalizability, as seen in CNN-based PIV methods.

1.5 Neural Optical Flow

We present *neural optical flow* (NOF), a framework utilizing a neural-implicit basis to model flow fields and facilitate many advantageous ML capabilities into the PIV pipeline. NOF also incorporates a differentiable, nonlinear image-warping operator to estimate large displacements without the need for a multi-resolution scheme, greatly reducing hyper-parameter counts. The velocity field in world space is represented as a continuous function of space and time, making the method naturally extensible to higher-dimensional techniques like stereo or tomographic PIV. This space–time formulation offers substantial compression, reduces computational costs, and ensures seamless handling of steady and unsteady flows.

Our framework leverages the strengths of Qi et al.'s neural-implicit OF method while eliminating the need for HDNet to enforce mass continuity. This is achieved by reparameterizing the velocity field in terms of a scalar or vector potential, from which velocity can be constrained to solenoidal or irrotational fields. Additionally, we can incorporate the Navier–Stokes equations into the optimization process, transforming NOF into a physics-informed neural network (PINN) [44]. By incorporating governing equations, our method operates as a PINN-based data assimilation algorithm, providing continuous, analytical representations of flow variables (e.g., velocity, pressure) that are trained to obey physical laws. Physics or regularization is achieved by minimizing equation residuals, improving flow state reconstruction from sparse or poor quality measurements and enabling direct pressure inference from PIV images [53].

In addition to supporting data assimilation, NOF natively integrates stereo PIV by modeling fields in a physical space and employing forward and inverse camera transforms to project between sensor and world coordinates. This allows us to use a single, continuous velocity field across both perspectives, ensuring consistent regularization and reducing the errors associated with the traditional recombination method of stereo PIV. NOF improves upon previous PINN-based methods, such as that of Cai et al. [53], by moving beyond the linear OF equation, making our approach more effective for turbulent flows with a broad range of spectral content and large displacements.

By incorporating aspects of both Qi et al. [52] and Cai et al. [53], with our own innovations, we introduce an accurate and robust OF technique capable of state-of-the-art performance. We validate NOF on both synthetic and experimental PIV datasets, comparing its performance to conventional CC and a state-of-the-art wavelet-based OF method. Our results demonstrate NOF's superior accuracy, robustness, and computational efficiency across a range of scenarios. The remainder of this paper is organized as follows: Chapter 2 introduces the concepts underlying NOF. In Chapter 3, we describe the synthetic and experimental flow cases used for evaluation. Numerical and experimental results are then presented in Chapters 4 and 5, respectively, followed by our conclusions.

Chapter 2

Methodology

Neural optical flow estimates the underlying velocity field that warps the first image in a pair such that is matches the second image. In this short preamble, we introduce the coordinate transformations between the world and sensor spaces that will be fundamental to the underlying principles of NOF. The 2D plane of the laser sheet, $\mathcal{P}$ is comprised of a set of 3D points in world space, $\mathbf{x} = (x_1, x_2, x_3) \in \mathcal{P}$. The light scattered off of particles in the laser sheet are resolved into images in the sensor space, $\mathcal{S}$, where $\mathbf{s} = (s_1, s_2) \in \mathcal{S}$ represents a position on the sensor in pixel units. There exists a unique mapping between $\mathcal{P}$ and $\mathcal{S}$ in both directions as both are planar domains. The forward mapping, or "camera transform," is defined as

$$\Psi : \mathcal{P} \to \mathcal{S}, \tag{2.1}$$

and the inverse mapping, or inverse camera transform, is

$$\Psi^{-1} : \mathcal{S} \to \mathcal{P}. \tag{2.2}$$

Examples of Ψ and Ψ^{-1} relevant to PIV are provided in App. A. Both Ψ and Ψ^{-1} are projections through the lens of the camera, with the "pinhole" camera model. With the magnification of the optical system for light from a point $\mathbf{x}$,

$$M = \frac{f}{|\mathbf{x} - \mathbf{c}| - f} \quad , \tag{2.3}$$

where $\mathbf{c}$ denotes the camera's position and f is the focal length of the lens, and the pixel pitch, ψ, defined as the reciprocal of the pixel size, we can describe the estimation of velocity fields from PIV data.

In the following sections, we introduce a differentiable image-warping operator providing the basis of our technique and demonstrate its use in data losses for planar and stereo PIV. We then explore various regularization techniques, including both physics-*inspired* and physics-*informed* methods.

2.1 Neural Optical Flow

Optical flow is a computer vision method used to estimate the *apparent* motion (what can be thought of as a sensor-domain motion) in a sequence of images based on temporal changes in intensity. Over the years, many variations of OF have been developed, leveraging pixel-based methods, wavelets, or neural networks, as reviewed by Mendes et al. [54]. Our NOF technique builds on these approaches and is different from them by employing a neural-implicit representation of the velocity field in physical space, rather than sensor space. Like other OF algorithms, NOF assumes that the brightness of a displaced feature is conserved between consecutive frames:

$$I(\mathbf{s}, t) = I(\mathbf{s} + \Delta\mathbf{s}, t + \Delta t)\,. \tag{2.4}$$

Here, I is the image intensity at a sensor position $\mathbf{s}$ and time t; $\Delta\mathbf{s}$ and Δt are the image displacement and time interval between frames, respectively.

In PIV, the displacement of a visible feature (usually a bright spot) in the image corresponds to the advection of a particle by the fluid in world space,

$$\Delta \mathbf{x} = \int_t^{t+\Delta t} \underbrace{\frac{\mathrm{d}\widetilde{\mathbf{x}}(\tau)}{\mathrm{d}\tau}}_{\mathbf{v}(\widetilde{\mathbf{x}},\tau)} \mathrm{d}\tau, \tag{2.5a}$$

where $\widetilde{\mathbf{x}}$ is the streamline starting from position $\mathbf{x}$ at time t and $\mathbf{v} = (v_1, v_2, v_3)$ is the fluid velocity. The physical displacement is then projected onto the camera sensor,

$$\Delta \mathbf{s} \equiv \underbrace{\Psi(\mathbf{x} + \Delta \mathbf{x})}_{\mathbf{s}+\Delta\mathbf{s}} - \underbrace{\Psi(\mathbf{x})}_{\mathbf{s}}. \tag{2.5b}$$

Traditional OF aims to solve for the apparent displacement, $\Delta\mathbf{s}$, which evolves the image intensity I from time t to $t + \Delta t$. Once $\Delta\mathbf{s}$ is determined, it is mapped back to the two in-plane displacement components, $(\Delta x_1, \Delta x_2)$.[1] The fluid velocity is then approximated as $\mathbf{v} \approx \Delta\mathbf{x}/\Delta t$.

Most traditional OF techniques do not solve for $\Delta\mathbf{s}$ using Eq. (2.4), as it is a challenging nonlinear optimization problem. Rather than the nonlinear form, a common simplification is to approximate the right-hand side with a first-order Taylor series expansion, evaluated at $(\mathbf{s}, t)$,

$$I \approx I + \frac{\mathrm{d}I}{\mathrm{d}t}\Delta t \tag{2.6a}$$

$$= I + \left(\frac{\Delta \mathbf{s}}{\Delta t} \cdot \nabla_{\mathbf{s}} I + \frac{\partial I}{\partial t}\right)\Delta t, \tag{2.6b}$$

where $\partial I/\partial t$ is approximated by $[I(\mathbf{x}, t + \Delta t) - I(\mathbf{x}, t)]/\Delta t$ and $\nabla_{\mathbf{s}}$ is the gradient operator in sensor coordinates. As $\Delta t \to 0$, we get the linear OF equation,

$$\frac{\partial I}{\partial t} + \mathbf{v}_{\mathcal{S}} \cdot \nabla_{\mathbf{s}} I = 0, \tag{2.7}$$

where $\mathbf{v}_{\mathcal{S}}$ is the apparent velocity in the image, which can be scaled by $(\psi M)^{-1}$ to estimate $\mathbf{v}$. While this linearization works well for small displacements, it breaks down as $\Delta\mathbf{s}$ or Δt becomes large and requires the use of a multi-resolution scheme [19, 22, 55]. A robust motion estimation algorithm must effectively handle large displacements, so we adopt the generalized OF model from Eq. (2.4) for this work.

Motion estimation by OF is often formulated as an optimization problem with a data loss term, expressed either in integral form,

$$\mathcal{J}_{\text{data}} = \frac{1}{|\mathcal{S}|}\int_{\mathcal{S}} [I(\mathbf{s}, t) - I(\mathbf{s} + \Delta\mathbf{s}, t + \Delta t)]^2 \, \mathrm{d}\mathbf{s}, \tag{2.8a}$$

or differential form,

$$\mathcal{J}_{\text{data}} = \frac{1}{|\mathcal{S}|}\int_{\mathcal{S}} \left(\frac{\partial I}{\partial t} + \frac{\Delta \mathbf{s}}{\Delta t} \cdot \nabla_{\mathbf{s}} I\right)^2 \mathrm{d}\mathbf{s}. \tag{2.8b}$$

[1] This assumes the laser sheet aligns with the (x_1, x_2) coordinates, otherwise a coordinate transformation is required.

The latter assumes the linearization in Eq. (2.6). Regradless, the objective is to find the displacement $\Delta\mathbf{s}$ that minimizes $\mathcal{J}_{\text{data}}$. This is an ill-posed inverse problem with no unique solution, having only one piece of information from each pixel (i.e., the intensity change) and two unknown parameters, $(\Delta s_1, \Delta s_2)$: an issue known as the "aperture problem" [56]. This is also apparent in the inability to measure velocity components not perpendicular to the isocontours of intensity. To ensure a unique, physically plausible solution, most OF techniques introduce an explicit penalty term, $\mathcal{J}_{\text{penalty}}$, which promotes desirable characteristics, such as spatial smoothness, in the solution. The total loss function is a weighted sum of the data and penalty terms,

$$\mathcal{J}_{\text{total}} = \mathcal{J}_{\text{data}} + \lambda\mathcal{J}_{\text{penalty}} \tag{2.9}$$

The weighting parameter λ must be carefully selected to balance the losses and is typically chosen heuristically. The optimal displacement field, $\Delta\mathbf{s}^*$, is obtained by minimizing the total loss,

$$\Delta\mathbf{s}^* = \arg\min_{\Delta\mathbf{s}} \mathcal{J}_{\text{total}} \,. \tag{2.10}$$

Neural optical flow is distinguished by its use of a neural-implicit representation of the velocity field. Rather than relying on pixel- or wavelet-based methods, a neural network, $\mathcal{N}$, models the fluid velocity as a continuous function of space and time,

$$\mathcal{N} : (\mathbf{x}, t) \mapsto \mathbf{v}. \tag{2.11}$$

In other words, $\mathcal{N}$ is a network that takes $\mathbf{x}$ and t as inputs and outputs $\mathbf{v}$. Additional variables like pressure can also be outputted, as demonstrated in Sec. 4.3. Details of the network architecture are provided in B. Given an estimated velocity field from $\mathcal{N}$, the following sequence is used to compute the displacement and objective loss:

1. **Compute physical displacement:** A displacement $\Delta\mathbf{x}$ is obtained by numerically integrating the velocity field $\mathbf{v}$ using Eq. (2.5a).
2. **Transform to sensor space:** The physical displacement is projected onto the sensor as $\Delta\mathbf{s}$ via Eq. (2.5b).
3. **Evaluate intensities for data loss:** The intensities $I(\mathbf{s}, t)$ and $I(\mathbf{s} + \Delta\mathbf{s}, t + \Delta t)$ are sampled for use in the integral-form data loss, Eq. (2.8a).

The following section details and visually presents these steps.

The NOF framework offers several advantages over existing OF methods. First, numerical integration of the velocity field allows it to handle large displacements without requiring a multi-resolution scheme, which would introduce many hyper-parameters [57, 58, 23, 38, 39]. Second, NOF extends naturally to stereo PIV, as discussed in Sec. 2.3: the neural network represents all three velocity components in physical space and optimizes across both imaging perspectives simultaneously, unlike traditional methods which solve for the apparent motion from different cameras. Third, the method leverages a C^∞ neural representation, enabling continuous regularization through automatic differentiation, as outlined in Sec. 2.4. Implicit regularization is also achieved by tuning the spectral content embedded in the network's Fourier encoding, per B, in many cases removing the need for explicit regularization. Lastly, both weak- and hard-form physics-based constraints are introduced to refine velocity estimates and infer additional flow states, such as pressure, which is described in Sec. 2.5.

2.2 Data Loss for Planar PIV

A key element of any OF technique is the warping operator, as the quality of the velocity field is assessed by warping the first image to approximate the second one. An ideal warping operator, $\mathcal{W}$, can be defined in terms of the sets

$$\mathcal{I}(t) = \{I(\mathbf{s}, t) \mid \mathbf{s} \in \mathcal{S}\} \quad \text{and} \quad \delta\mathcal{S} = \{\Delta\mathbf{s}(\mathbf{s}) \mid \mathbf{s} \in \mathcal{S}\}, \tag{2.12}$$

where $\mathcal{I}$ represents the imaged intensity field at time t across the sensor domain $\mathcal{S}$, and $\delta\mathcal{S}$ is the deflection field. The warping operator acts on this set as

$$\mathcal{W} : (\mathcal{I}, \delta\mathcal{S}) \mapsto \mathcal{I}', \tag{2.13}$$

where $\mathcal{I}'$ is the image at time t warped by the displacement field $\delta\mathcal{S}$.

The linear OF warping operator, commonly used in OF algorithms, employs Eq. (2.7) to describe changes to the initial image. Discrete derivatives of I, evaluated at the pixel centroids, are combined with the estimated flow state to generate the intensity change at each pixel and thus the warped image. This warped image is then compared to the experimental data, forming the objective loss in Eq. (2.8b).

The nonlinear warping operators used in conjunction with Eq. (2.8a) do not rely on such approximations. Nonlinear warping can be categorized into two classes. The first, more general approach advects the intensity at each position $\mathbf{s}$ to $\mathbf{s} + \Delta\mathbf{s}$, with the warped image $\mathcal{I}'$ comprising the sum of deposited intensities at these new locations. However, this formulation is non-injective and not invertible, as illustrated by the scenario where two points in the first frame advect to the same location in the second frame. If we were to have a bright particle and a dim particle advect to the same position (possible due to out-of-plane effects) the final position would likely not have the same intensity as either of the initial particles, and thus our assumption of intensity conservation is broken.

Furthermore, many optimization schemes call for an invertible mapping. This second class of warping assumes injectivity, ensuring that the operator satisfies

$$\mathcal{W}(\mathcal{I}, \delta\mathcal{S}) = \mathcal{I}' \quad \text{and} \quad \mathcal{I} = \mathcal{W}(\mathcal{I}', -\delta\mathcal{S}). \tag{2.14}$$

With the added assumption of surjectivity, every point $\mathbf{s} \in \mathcal{S}$ in the first frame maps to a unique point in the second frame, allowing the integrals in Eq. (2.8a) to be approximated to arbitrary precision. In this work, we adopt this assumption, implying that all positions in $\mathcal{S}$ are advected to distinct locations on the sensor in the second frame, although there are no rigorous constraints and our method remains robust even when these assumptions are not strictly met.

Our warping scheme involves five steps, illustrated in Fig. 2.1. First, a set of points is uniformly sampled from the sensor domain, $\mathcal{S}$. These points, $\{\mathbf{s}_1, \mathbf{s}_2, \dots\}$, are then mapped to world coordinates using the inverse camera transform, $\mathbf{x}_i = \Psi^{-1}(\mathbf{s}_i)$, as detailed in A. Next, the points are advected from $\mathbf{x}_i$ to $\mathbf{x}_i + \Delta\mathbf{x}_i$ using a forward Euler scheme to approximate the integral in Eq. (2.5a). Numerical integration of the continuous velocity field is crucial for managing large displacements; we use a time step of $\Delta t/5$, which proved stable across all tested flow scenarios, unless otherwise noted. Testing with a step size of Δt showed inferior accuracy compared to our multi-step approach, except for largely linear flows that did not show a large improvement. While

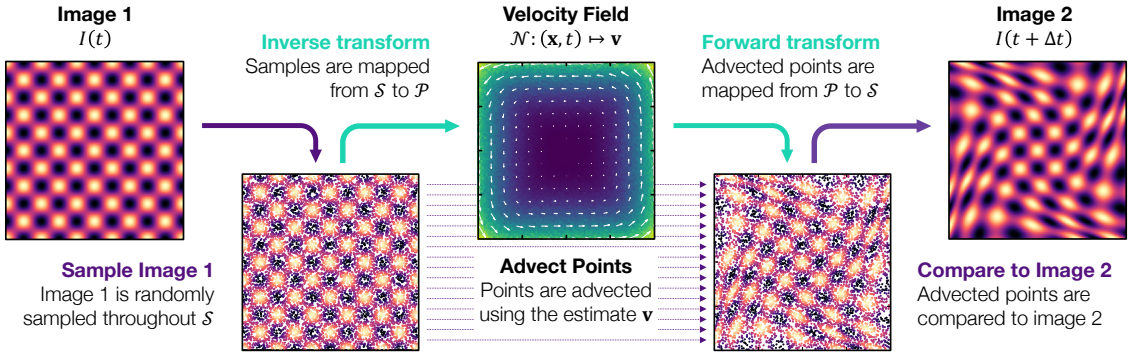

Figure 2.1: Schematic of the image warping procedure in NOF. Points are sampled in $I(t)$, advected to $t + \Delta t$ by integrating $\mathbf{v}$, and then compared to the corresponding intensity in $I(t + \Delta t)$.

this study uses a forward Euler scheme, future work may explore higher-order integration methods for increased computational efficiency.

In the fourth step, the final locations of the points are transformed back to sensor coordinates using the camera transform,

$$\mathbf{s}_i + \Delta\mathbf{s}_i = \Psi(\mathbf{x}_i + \Delta\mathbf{x}_i)\,. \tag{2.15}$$

Next, pixel intensities are sampled from the first image at $\mathbf{s}_i$ and from the second image at $\mathbf{s}_i + \Delta\mathbf{s}_i$. When the displacement field is accurate, these intensity values should match, assuming our injective principle holds. Intensities are retrieved via interpolation of the discrete image data: $\mathbf{I} = \{I_{i,j} \mid i = 1, \ldots m, j = 1, \ldots n\}$, where $I_{i,j}$ is the intensity of the (i, j)th pixel in an $m \times n$ image. While both bilinear and bicubic interpolation schemes were tested, bilinear interpolation is used throughout this work due to its lower computational cost and comparable accuracy across all cases. The interpolation operator is defined as

$$\mathcal{B} : [\mathbf{I}(t)\,, \mathbf{s}] \mapsto \hat{I}(\mathbf{s}, t)\,, \tag{2.16}$$

where $\hat{I}$ is the interpolated estimate of the continuous intensity field, I. Further details about the interpolation scheme are provided in C.

Lastly, in the fifth step, we use Monte Carlo sampling to evaluate an integral-form data loss,

$$\mathcal{J}^k_{\text{data}} = \frac{1}{|\mathcal{S}_k|}\int_{\mathcal{S}_k} \{\mathcal{B}[\mathbf{I}(t)\,, \mathbf{s}] - \mathcal{B}[\mathbf{I}(t + \Delta t)\,, \mathbf{s} + \Delta\mathbf{s}_k]\}^2\,\mathrm{d}\mathbf{s},$$

where k denotes the camera index (to allow for multi-camera setups),

$$\Delta\mathbf{s}_k = \Psi_k\Big[\Psi_k^{-1}(\mathbf{s}) + \Delta\mathbf{x}_k\Big] - \mathbf{s},$$

and $\Delta\mathbf{x}_k$ is the solution to Eq. (2.5a) initialized at $\Psi_k^{-1}(\mathbf{s})$. For planar imaging, only a single camera is used, and the camera index can be omitted. However, capturing out-of-plane velocity components requires multiple cameras or auxiliary devices, such as endoscopes or fibers. In such cases, a separate data loss must be computed for each image, as discussed in the next section.

2.3 Data Loss for Stereo PIV

In most realistic flow scenarios, including turbulent flows, vortex-dominated flows, unstable shear flows, and more, 3 dimensional effects are severe. Stereoscopic PIV (stereo PIV) uses two cameras to simultaneously capture particles within the measurement plane from distinct perspectives. The information from these different perspectives allows for the extraction of out-of-plane fluid motion and helps to reduce perspective errors in the planar motion [59]; operating conceptually similar to the way our eyes extract depth information. This velocimetry technique produces two-dimensional, three-component (2D3C) velocity fields defined on a plane but comprising of three component velocity vectors that have become essential for accurately measuring velocity in flows with significant vortical content or other out-of-plane motions.

Most stereo PIV setups follow a translational or rotational configuration [60]. Figure 2.2 shows a visual illustration of planar and stereo PIV in different configurations. In the translational version, both cameras have parallel optical axes that are perpendicular to the laser sheet, i.e., pointing directly forward. This setup is easy to implement, provides uniform magnification across the region of interest, and ensures a sharp focus, similar to single-camera planar PIV. However, the cameras in this configuration have a limited "common area" (i.e., the region of the laser sheet visible to both cameras). Moving the two cameras closer increases this area, but sacrifices the accuracy of estimating out-of-plane motion. Translating the sensors relative to the lenses can also increase the common area, but is challenging with many commercial cameras and increases lens vignetting as the sensors move off-axis, especially at the large offsets needed for accurate out-of-plane motion estimation.

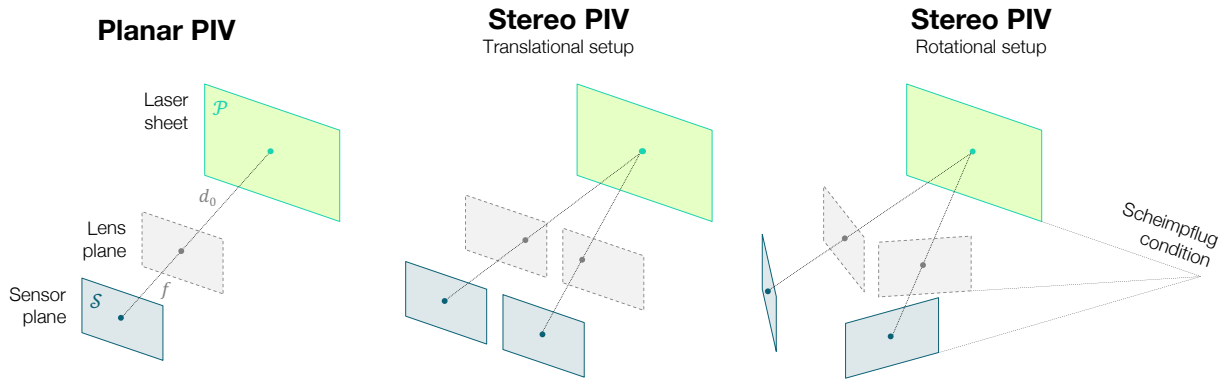

Figure 2.2: PIV setups: (left) planar, (middle) translational stereo, and (right) rotational stereo. Green planes are the laser sheet, cobalt planes are the sensors, and gray planes are the lenses. Diagram is not to scale.

The rotational setup addresses these challenges by turning both cameras to point them towards the center of the region of interest. Scheimpflug adapters, which shift the sensor and lens planes such that they are not parallel to one another, are typically used to maintain a sharp focus from these oblique angles; without this modification, each camera's plane of focus would not coincide with the laser sheet. This configuration overcomes the off-axis angle limitations of the translational method, reducing vignetting and enabling accurate out-of-plane motion estimation over a larger area. However, rotational setups introduce non-uniform magnification across the field of view,

which complicates camera calibration, data processing, and the computational framework for the camera transforms. Both configurations are discussed in detail by Prasad [60].

In this work, we demonstrate NOF using the translational configuration for simplicity. However, our algorithm is equally applicable to the rotational configuration with appropriate adjustments to the camera transform. Details of the translational and rotational transforms are provided in A.4.

Traditional stereo PIV processing performs an analysis on the images from each perspective separately using OF, CC, or a similar technique. These fields are then mapped onto the laser plane in world coordinates and stitched together, enabling the estimation of both in-plane and out-of-plane motion, as described by Eq. (A.15) in A.4. In contrast, NOF offers the ability to performs the inversion directly in world space on a unified velocity field between all perspectives. Our stereo data loss follows the same structure as Eq. (2.17), with points sampled on the sensors, transformed into world coordinates, advected, and then mapped back to image space. The key difference is that stereo NOF forms a composite data loss from both perspectives, with one component per camera, using their respective forward and inverse camera transforms:

$$\mathcal{J}_{\text{data}} = \mathcal{J}^{\text{L}}_{\text{data}} + \mathcal{J}^{\text{R}}_{\text{data}}. \tag{2.18}$$

Here, $\mathcal{J}^{\text{L}}_{\text{data}}$ and $\mathcal{J}^{\text{R}}_{\text{data}}$ are losses for the left and right cameras, respectively, which have different Ψ and Ψ^{-1} operators.

Although our measurement is planar, advection by the 2D3C velocity field can move points out of the plane/laser sheet, as such the network's spatial input must be 3D, unlike planar NOF which only inputs the 2D coordinates of the laser sheet. In principle then, sampling of the network $\mathcal{N}$ could be performed throughout a thin volume surrounding the laser sheet, within which particles are illuminated by a Gaussian laser slab of finite thickness. However, in our tests, sampling within the central $x_1{-}x_2$ plane yielded the best results. Future work will explore improved sampling schemes to more accurately capture out-of-plane effects.

2.4 Physics-Inspired Regularization

An explicit penalty term is often included in the aggregate loss to close the OF equations and promote desirable solution characteristics. Most OF regularization methods use "physics-inspired" penalties that do not strictly align with the governing equations and thus are not strictly physically realistic but encourage properties like smoothness or a known correlation length scale [61]. NOF can apply the same penalty terms as other OF algorithms for PIV but does so in a *truly* continuous form. For comparison, WOF relies on gradient operators within the discrete signal space associated with the wavelet transform, making them finite difference stencils. Moreover, we regularize the flow velocity within the measurement plane in world space, $\mathcal{P}$, rather than sensor space; this ensures consistent regularization for stereo PIV and facilitates the more comprehensive, physics-based penalties discussed in the next section.

A common penalty choice is the *first-order* Tikhonov regularization method introduced to OF by Horn and Schunck [57],

$$\mathcal{J}_{\text{penalty}} = \frac{1}{|\mathcal{P}|} \int_{\mathcal{P}} \|\nabla_{\mathbf{x}} \mathbf{v}\|_2^2 \, \mathrm{d}\mathbf{x}, \tag{2.19}$$

where $\|\cdot\|_2$ is the Euclidean norm. Tikhonov regularization promotes *globally* smooth solutions in the L^2 sense. Another common choice is total variation (TV) regularization, which minimizes the overall gradient content to promote *piecewise* smooth solutions that can sustain sharp discontinuities, i.e., enforcing smoothness in the L^1 sense:

$$\mathcal{J}_{\text{penalty}} = \frac{1}{|\mathcal{P}|} \int_{\mathcal{P}} \|\nabla_{\mathbf{x}} \mathbf{v}\|_1 \, \mathrm{d}\mathbf{x}. \tag{2.20}$$

Note that $\|\cdot\|_1$ is the Manhattan norm, which is non-differentiable and typically requires a specialized optimization technique. Higher-order regularization terms, such as

$$\mathcal{J}_{\text{penalty}} = \frac{1}{|\mathcal{P}|} \int_{\mathcal{P}} \left(\frac{\partial^3 v_1}{\partial x_1^3} \right)^2 + \left(\frac{\partial^3 v_2}{\partial x_2^3} \right)^2 \mathrm{d}\mathbf{x} \tag{2.21}$$

can improve velocity estimation without excessively smoothing the field or penalizing divergence and curl [12, 27, 28]. A particularly effective approach for 2D velocity fields is div–curl regularization,

$$\mathcal{J}_{\text{penalty}} = \frac{1}{|\mathcal{P}|} \int_{\mathcal{P}} \|\nabla_{\mathbf{x}} (\nabla_{\mathbf{x}} \cdot \mathbf{v})\|_2^2 + \|\nabla_{\mathbf{x}} (\nabla_{\mathbf{x}} \times \mathbf{v})\|_2^2 \, \mathrm{d}\mathbf{x}, \tag{2.22}$$

which aims to smooth the flow by penalizing gradients of divergence and curl while preserving the structure of vortical and compressive regions [58, 23, 62, 28].

Penalizing higher-order derivatives in an explicit penalty term often introduces numerical sensitivities that challenge optimization algorithms and severely increase computational costs. In NOF, the network architecture provides implicit regularization, as outlined in B. By adjusting the Fourier encoding, the spectral content that the neural network is capable of learning can be regulated. Thus, the bandwidth of the network may be tailored to the flow, e.g., based on an estimated Reynolds number or known spectral properties [63]. Systematic testing demonstrated that implicit regularization is both more accurate and efficient than explicit methods in NOF. Therefore, we employ implicit regularization throughout this work unless we state otherwise.

2.5 Physics-Based Constraints

Explicit physics-based constraints enhance the accuracy and physical consistency of solutions by penalizing or constraining residuals of underlying, physical equations. While some constraints can be applied in other OF methods, such as mass continuity for incompressible flows [23, 64, 65], NOF offers distinct advantages. Specifically, our approach enables time-resolved reconstructions that include the Navier–Stokes equations. Additionally, NOF's space–time representation allows us to exploit sparsity in the time domain: an advantage not utilized by existing OF methods.

Many flows of interest are incompressible, making it advantageous to extract the velocity field from a potential field in such a way that it inherently satisfies the continuity equation,

$$\frac{\partial \rho}{\partial t} + \nabla_{\mathbf{x}} \cdot (\rho \mathbf{v}) = 0 \Rightarrow \nabla_{\mathbf{x}} \cdot \mathbf{v} = 0. \tag{2.23}$$

This is straightforward to implement in NOF. For example, a 2D divergence-free velocity field can be represented using a scalar potential, φ, where $\mathcal{N} : (\mathbf{x}, t) \mapsto \varphi$ and

$$\mathbf{v} = \left(-\frac{\partial \varphi}{\partial y}, \frac{\partial \varphi}{\partial x} \right). \tag{2.24}$$

Taking the divergence of this field confirms compliance with the continuity equation,

$$\nabla_{\mathbf{x}} \cdot \mathbf{v} = -\frac{\partial^2 \varphi}{\partial x \partial y} + \frac{\partial^2 \varphi}{\partial y \partial x} = 0. \tag{2.25}$$

Although this parametrization is limited to 2D flows, continuity can still be embedded as a hard constraint in 3D settings via a vector potential field,

$$\boldsymbol{\varphi} = (\varphi_1, \varphi_2, \varphi_3)\,. \tag{2.26}$$

The network maps $(\mathbf{x}, t)$ to $\boldsymbol{\varphi}$ and the velocity field is given by

$$\mathbf{v} = \nabla_{\mathbf{x}} \times \boldsymbol{\varphi}, \tag{2.27}$$

ensuring, again, that the velocity field remains divergence-free,

$$\nabla_{\mathbf{x}} \cdot (\nabla_{\mathbf{x}} \times \boldsymbol{\varphi}) = 0. \tag{2.28}$$

In practice, two components of $\boldsymbol{\varphi}$ may be sufficient, with $\varphi_3 = 0$. Further research is needed to assess this approach for 3D3C measurements, such as tomographic PIV [5] and Lagrangian particle tracking [17]. From current testing, it appears the required out-of-plane gradients of $\boldsymbol{\varphi}$ nullify its use in 2D3C measurements. It is also possible to constrain the network to irrotaional solutions in planar and 3D domains, in similar ways as shown above.

In addition to hard constraints, we incorporate soft constraints by including the residuals of the 2D Navier–Stokes equations to enable *direct* time-resolved pressure estimation from PIV data for 2D flows:

$$\mathcal{J}_{\text{penalty}} = \frac{1}{|\mathcal{P} \times \mathcal{T}|} \int_{\mathcal{T}} \int_{\mathcal{P}} \underbrace{\left(\nabla_{\mathbf{x}} \cdot \widehat{\mathbf{v}}\right)^2}_{\text{continuity}} + \underbrace{\left\| \frac{\partial \widehat{\mathbf{v}}}{\partial t} + \widehat{\mathbf{v}} \cdot \nabla_{\mathbf{x}} \widehat{\mathbf{v}} + \nabla_{\mathbf{x}} \widehat{p} - \frac{1}{Re} \nabla^2 \widehat{\mathbf{v}} \right\|_2^2}_{\text{momentum conservation}} \mathrm{d}\mathbf{x}\, \mathrm{d}t. \tag{2.29}$$

In this expression, $\mathcal{T}$ is the time domain of the experiment, while $\widehat{\mathbf{v}} = \mathbf{v}/v_0$ and $\widehat{p} = p/(\rho_0 u_0^2)$ represent the dimensionless velocity and pressure, respectively, and v_0, ρ_0, and ℓ_0 are the characteristic velocity, density, and length scales. The dimensionless Navier–Stokes and mass continuity equations are used to ensure that their magnitudes are similar to one another. The reference Reynolds number is

$$Re = \frac{\rho_0 u_0 \ell_0}{\mu}. \tag{2.30}$$

Note that the mass continuity residuals in Eq. (2.29) are redundant when using one of the divergence–free hard constraints presented above. Incorporating governing equation residuals into the loss effectively transforms NOF into a PINN, where the estimated velocity field not only aligns with PIV data but also approximately solves the Navier–Stokes equations. This enhances the estimation of velocity, especially in regions of sparse information, and enables simultaneous pressure inference. However, it only applies to 2D flows, as planar cross-sections of fully-3D flows do not satisfy the 2D equations. Usage of the 3D Navier–Stokes equations is hampered by the massive uncertainty in out-of-plane gradients. In such cases, fully 3D measurements may be required for effective data assimilation [66, 67].

Chapter 3

Flow Scenarios and Metrics

NOF is evaluated across a variety of synthetic and experimental test cases. The synthetic cases provide idealized and noise-free conditions to gauge accuracy under varying seeding densities, robustness against different flow scenarios, and the effects of out-of-plane motion using both 2D and 3D flows. The experimental cases offer a test of NOF's performance under conditions with noise, shot-to-shot laser intensity variations, and non-uniform seeding and illumination, all of which are common in experimental scenarios. Sample images for each PIV scenario are shown in Fig. 3.1, with detailed descriptions provided in the following sections.

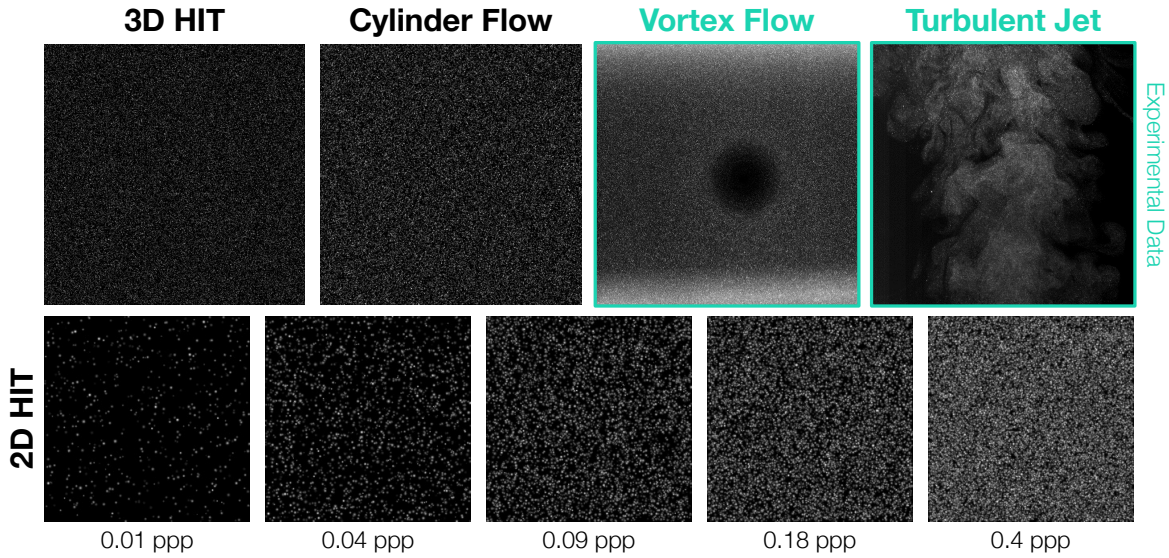

Figure 3.1: Sample particle images from synthetic and experimental flow cases. Experimental cases from previous PIV challenges [1, 2] are highlighted with a teal border.

The availability of ground truth data in synthetic testing provides a key advantage, allowing for precise error quantification. We report normalized root-mean-square errors (NRMSEs) of the fluctuating components as an error metric. This allows us to assess whether each technique can accurately capture unsteady flow behavior, as the mean components are comparatively easy to estimate. For a quantity of interest g, we apply a Reynolds decomposition, $g = \overline{g} + g'$. The NRMSE for g is

$$e_g = \frac{\left\langle \left\| g' - g'_{\text{exact}} \right\|_2 \right\rangle}{\left\langle \left\| g'_{\text{exact}} \right\|_2 \right\rangle} \tag{3.1}$$

where g'_{exact} is the exact fluctuating component and $\langle \cdot \rangle$ indicates a spatial average over the measurement domain,

$$\langle g \rangle = \frac{1}{|\mathcal{P}|} \int_{\mathcal{P}} g(\mathbf{x}) \, \mathrm{d}\mathbf{x}. \tag{3.2}$$

In our Reynolds decomposition, $\overline{g}$ is the time average of g over the measurement interval. For forced isotropic turbulence, there is no mean value.

3.1 Synthetic PIV Cases

We use three DNS datasets for synthetic testing of NOF's capacity for PIV analyses: 2D homogeneous isotropic turbulence (HIT) from Carlier and Wieneke [68], 3D HIT from the Johns Hopkins Turbulence Database [69], and a 3D cylinder wake flow from Raissi et al. [44]. These cases, referred to as 2D HIT, 3D HIT, and cylinder flow, contain 100, 51, and 201 frames, respectively.

Table 3.1: Parameters and Dimensions of the DNS Flow Cases

Flow case	Re	**Domain size**		**Time step**		**Grid**
		DNS	dim., mm	DNS	dim., ms	
2D HIT	3000	$2\pi \times 2\pi$	100×100	0.01	0.845	256×256
3D HIT	5405	$2\pi \times 2\pi \times 5\pi/64$	$150 \times 150 \times 5.9$	0.002	0.211	$1024 \times 1024 \times 40$
Cylinder	100	$5 \times 5 \times 5$	5×5	0.08	0.80	$51 \times 51 \times 51$

Each dataset is dimensionalized to simulate realistic flow imaging conditions, assuming the fluid is water. The hypothetical setups are tailored to match standard PIV configurations. Table 3.1 summarizes key dimensionless and dimensional quantities for each scenario, with Reynolds numbers defined using the energy injection scale (2D and 3D HIT) or the cylinder diameter.

Synthetic particle images are generated by numerically advecting tracer particles and projecting their positions into sensor space. As described in A, we assume a pinhole camera model with a magnification of 1/3. Object distances are set to 120, 150, and 60 mm for the 2D HIT, 3D HIT, and cylinder cases, respectively, with focal lengths of 40, 50, and 20 mm. The image plane is coincident with the physical x_1-x_2 plane. For planar setups, the camera is centered at $x_1 = x_2 = 0$ mm. The inter-lens spacing is set to 75 mm in our stereo 3D HIT test. In all scenarios, the synthetic laser sheet has a Gaussian profile centered on $x_3 = 0$ mm, with a standard deviation in the x_3-direction of 0.63 mm.

The physical domain is uniformly seeded with particles, which are then advected according to the velocity field. For the 3D HIT and cylinder flows, a second-order Runge–Kutta scheme is used, while a forward Euler scheme with at least 50 steps per frame is applied to the 2D HIT flow. Periodic boundary conditions are applied to the HIT cases. For the cylinder flow, particles exiting the domain are re-injected at the opposite boundary at a randomized height. Particle locations are projected onto the image plane using Eq. (A.11a) for planar setups and Eq. (A.13) for stereo setups. Image resolutions are matched to the grid sizes listed in Table 3.1, except for the cylinder case where images of 841×841 px are synthesized. Particles are modeled as Gaussian blobs with a standard deviation of 1.25 pixels, giving them an apparent diameter of 3–4 pixels. As shown in Fig. 3.1, the particle density is varied from 0.01 to 0.4 ppp to assess the effects of seeding density in the 2D HIT case.

3.2 Synthetic BOS Case

In addition to our main concern of validating NOF for PIV analyses, we test NOF's deflection sensing capabilities upon a synthetic BOS dataset generated from DNS of buoyancy driven turbu-

lence on a 1024^3 grid available from the John Hopkins Turbulence Database [70]. This test case, referred to hereinafter as buoyancy driven turbulence or BDT, begins as a homogeneous arrangement of random blobs of two fluids at rest. The miscible two-fluid Navier–Stokes equations are then solved using a pseudo-spectral method. Turbulent motion arises as a result of body forces caused by a density difference in the two fluids and continues until the fluids are molecularly mixed and turbulence decays. The Froude number (Fr) and Schmidt number (Sc) are both 1, with an Atwood number (A) of 0.05.

Before any synthetic BOS data can be generated from the DNS, we must dimensionalize the dataset to mimic realistic imaging conditions. We first selected a 20° C air/oxygen mixture subject to a mean pressure gradient as our light/heavy fluid system. This mixture of fluids produces Sc = .88 and A = .0435, both of which slightly under-approximate the DNS. We then use the mean kinematic viscosity of the air/oxygen mixture along with the known $Fr = 1$ and maximum turbulent Reynolds number $Re_\mathrm{T} = 17765$ to calculate the required velocity, length, and time scales to produce dimensionalized fields. From these dimensional fields, a $12 \times 12 \times 4.7$ cm^3 subvolume is selected for imaging.

The density of the mixture ρ_mix at a given point is related to the mass fraction and density of each fluid as

$$\rho_\mathrm{mix} = \left(\frac{Y_1}{\rho_1} + \frac{Y_2}{\rho_2} \right)^{-1} \tag{3.3}$$

where Y_i and ρ_i are the mass fraction and density of the i^th fluid. From the density and mass fractions at a point, the refractive index can be retreived as:

$$n = 1 + \rho_\mathrm{mix}(Y_1 G_1 + Y_2 G_2) \tag{3.4}$$

where $G_1 = 2.26 \times 10^{-4}$ m^3 kg^{-1} and $G_2 = 1.89 \times 10^{-3}$ m^3 kg^{-1}. It should be noted that the refractive index of oxygen has been increased by an order of magnitude to generate measurable deflections.

To generate synthetic BOS images, a camera and background setup must be defined with realistic parameters. As such a sensor size of 1024×1024 px with a pixel pitch of 3.45 μm is used. A realistic focal length f of 35 mm and an aperture setting of $f/7.1$ were chosen and used to generate images, although we use the simpler "pinhole" model with $f = \infty$ for this test case. Depth of field effect are included via Monte Carlo sampling of the aperture for a given f value; however, these effects are non-existent in the limit of $f = \infty$. The sensor plane, background, and domain are aligned, with the distance from camera to the center of the flow domain being 1 m and the background plane being positioned an additional 1 m farther away, resulting in a field of view through the fluid domain of $\sim$ 10 cm. Imaging occurs at 40 Hz for 5 s.

With the camera setup defined, images and ground-truth deflection fields are generated using nonlinear ray tracing with a fourth-order Runge-Kutta scheme. Rays propagate linearly from the background to the flow domain and linearly from the flow domain to the camera, with refraction outside of the flow domain being neglected. This is an approximation of the more physically realistic nonlinear path integration, which drastically reduces computational cost while only slightly degrading the resulting images. Images were generated with a variety of background patterns; we selected a superimposed sinusoid pattern with 10 periods along each side of the background.

3.3 Experimental Cases

We also demonstrate NOF on two experimental datasets: a strong vortex flow from the first international PIV challenge [1] and a time-resolved jet flow from the third challenge [2]. These cases are referred to as vortex flow and turbulent jet flow, respectively.

The vortex flow dataset [1] consists of two consecutive frames capturing a vortex formed behind the wingtip of a transport aircraft (ALVAST half-model) in a landing configuration. The main air flow is perpendicular to the laser plane, with a velocity of 60 m s^{-1}, and the measurement plane is positioned 1.64 m behind the wingtip. Images are taken with a progressive scan camera featuring a 1024×1280 px chip and 6.7×6.7 μm pixels. We crop the images to 1024×1024 px, reducing the field of view from 140×170 mm^2 to 140×140 mm^2. The magnification is approximately 0.05. Since the frame rate is not specified, we follow the original PIV challenge paper and report results in pixels-per-frame (ppf). This dataset is characterized by strong intensity gradients, a non-uniform particle density, and particle size variations, which pose significant challenges for PIV algorithms, making it an excellent test of NOF's robustness.

The turbulent jet flow dataset [2] features a nitrogen jet with an exit velocity of approximately 30 m s^{-1}. The nozzle has a diameter of 5 mm, and the flow is seeded with oil-mist particles averaging 5 μm in diameter. The domain is illuminated with 527 nm light from a high-speed dual-head PIV laser (New Wave Pegasus) operating at 5 kHz and imaged with a high-speed camera (Photron APX-RS). The camera has a 512×512 px sensor, 17×17 μm pixels, and magnification of 0.3 for the PIV plane. Imaging is performed at 10 kHz using frame straddling to capture the output of both laser heads, which have an inter-frame time separation of 10 μs. This dataset was selected to evaluate NOF's ability to handle noise and inter-frame intensity variations, which are common complications in a PIV experiment.

Chapter 4

Synthetic Results

This section assesses NOF's performance by examining various synthetic flow cases and comparing it against the CC and WOF methods. Throughout these tests, we perform CC with a standard cascade over 64^2-, 32^2-, and 16^2-pixel interrogation windows with aggressive overlapping of 75%. WOF hyperparameters are *optimally* tuned for each case. Several NOF variants with unique regularization schemes and flow parameterizations are explored, with their performance initially tested on the 2D HIT case across varying seeding densities. Spectral error analysis offers deeper insight into each method's ability to capture different flow scales. Subsequently, we illustrate NOF's accuracy and adaptability in both planar and stereo PIV setups using 3D HIT data, which includes significant out-of-plane motion. Lastly, we highlight NOF's capability to directly recover pressure fields from planar PIV measurements in the cylinder flow case.

4.1 2D HIT

We begin with the results of the 2D HIT test. Figure 4.1 compares the ground truth DNS data to the reconstructed fields obtained from CC, WOF, and NOF. The vanilla NOF implementation, utilizing implicit regularization as outlined in Sec. 2.4, conducts single-shot reconstructions on a pair of particle images. These results were achieved at a seeding density of 0.17 ppp, a typical value in PIV experiments. This density avoids excessive particle overlap, as illustrated in Fig. 3.1, while providing adequate particle density to resolve the Taylor micro-scale of turbulence, as demonstrated in Fig. 4.2.

Qualitatively, both NOF and WOF achieve high accuracy in recovering the velocity field, closely resembling the ground truth, whereas CC introduces noticeable low-pass filtering. This observation is corroborated by the absolute error distributions shown at the bottom of Fig. 4.1, where NOF and WOF produce significantly lower errors compared to CC. However, WOF displays slightly more artifacts than NOF, especially near the boundaries. This is consistent with observations in the experimental jet case in Sec. 5.2, shown in Fig. 5.3. Quantitatively, NRMSEs for the velocity magnitudes are 17.3%, 6.2%, and 5.1% for CC, WOF, and NOF, respectively.

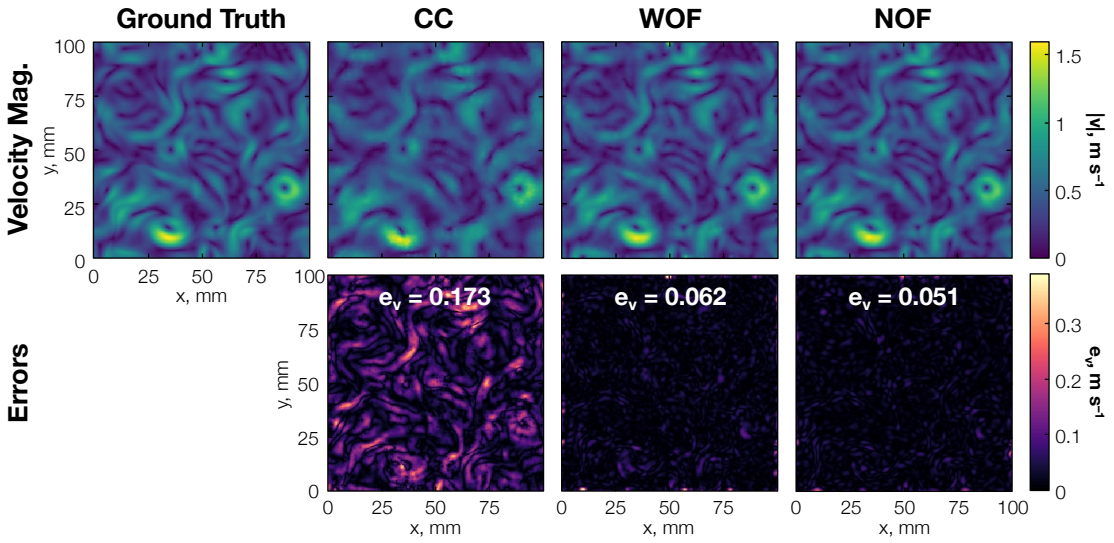

Figure 4.1: Exact and reconstructed 2D HIT velocity magnitude fields (top) and absolute pointwise errors (bottom). NRMSEs are superimposed on the absolute error plot.

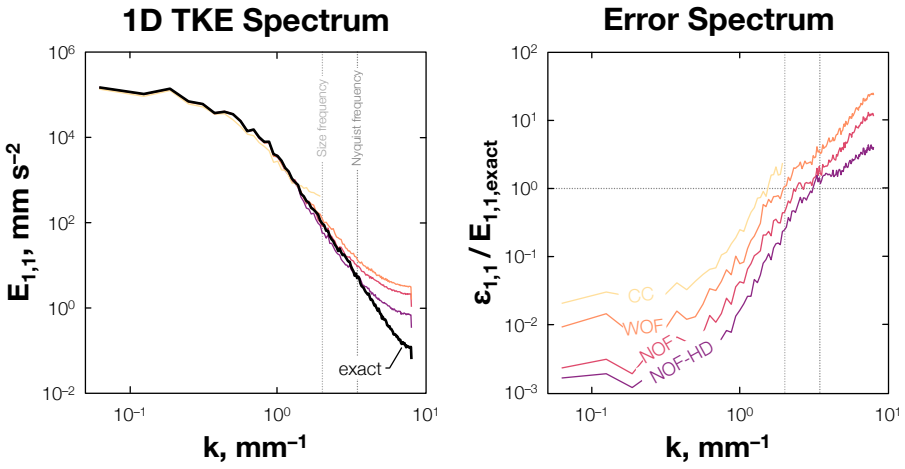

Figure 4.2: Comparison of 1D TKE spectra of flows reconstructed by different methods and exact DNS data (left) and normalized error spectra (right).

Although NRMSEs quantify the overall flow accuracy across the domain, spectral analysis provides a more detailed evaluation of each method's performance at varying length scales. Figure 4.2 presents 1D turbulent kinetic energy (TKE) spectra, $E_{1,1}$, for velocity fields reconstructed by CC, WOF, NOF, and NOF-HD. The NOF-HD variant incorporates a hard constraint to enforce mass continuity, as outlined in Sec. 2.5.

To interpret trends in the error spectra, we define the particle size and spacing-Nyquist frequencies,

$$\kappa_\mathrm{p} = \frac{2\pi}{d_\mathrm{p}} \quad \text{and} \quad \kappa_\mathrm{N} = \frac{\pi}{\ell_\mathrm{p}}, \tag{4.1}$$

respectively, where d_p is the apparent particle diameter on the sensor projected into physical space and ℓ_p is the mean inter-particle spacing. Specifically, κ_p represents the spatial frequency associated with the particle image size, below which flow features are obscured by the particle. Meanwhile, κ_N indicates the maximum recoverable frequency for a given seeding density, following the Shannon–Nyquist sampling theorem [71].

We propose that, without physics-based constraints, the accuracy of optical flow (OF) methods is inherently limited by the lowest of these frequencies, beyond which finer flow details remain unresolved. As illustrated on the left side of Fig. 4.2, CC diverges from the ground truth spectrum well before reaching the particle size frequency, constrained by the low cutoff of its large interrogation window. In contrast, WOF and NOF align closely with the ground truth spectrum up to the particle size frequency. Notably, NOF-HD exceeds this boundary, maintaining accuracy up to the Nyquist frequency. This demonstrates the advantage of integrating physical constraints into OF algorithms, allowing for improved resolution of fine-scale flow features.

Spectral errors for each method, normalized by the true kinetic energy of the flow at each wavenumber, $\varepsilon_{1,1}/E_{1,1,\mathrm{exact}}$, provide a detailed assessment of reconstruction accuracy across varying length scales. This metric, displayed on the right side of Fig. 4.2, highlights the superior performance of both NOF variants across the spectrum. All methods follow a similar pattern: low errors at small wavenumbers, followed by exponential growth at higher wavenumbers. At the particle size frequency, CC and WOF approach nearly 100% error, while NOF-HD maintains a significantly lower error of approximately 20%. Interestingly, both WOF and the standard NOF exhibit a

consistent exponential error increase at high wavenumbers. In contrast, NOF-HD shows a slower error growth, particularly beyond the Nyquist frequency, indicating that enforcing mass continuity reduces particle-induced artifacts and suppresses random high-frequency errors. Nonetheless, NOF-HD is unable to resolve flow behavior at frequencies exceeding κ_N.

Table 4.1: Errors for State-of-the-Art Motion Estimation Schemes

Regime	**Algorithm**					
	CC	WOF	NOF	NOF-HD	NOF-TR	NOF-phys
Sub-Nyquist	20.94	9.13	6.76	5.26	8.95	8.16
Super-Nyquist	N/A	188.42	134.78	114.12	141.47	106.22

To more thoroughly evaluate the accuracy of flow reconstruction across both resolved and unresolved scales (excluding the effects of finite particle image size), we compute *sub-* and *super-*Nyquist errors for the velocity fields, as detailed in Table 4.1. Sub-Nyquist errors are determined by low-pass filtering the velocity field in the Fourier domain with a cutoff frequency at κ_N, setting all Fourier components with $|\boldsymbol{\kappa}| \geq \kappa_N$ to zero. The filtered field is then reconstructed using the inverse Fourier transform, and sub-Nyquist errors are calculated according to Eq. (3.1). Super-Nyquist errors are similarly calculated by high-pass filtering, zeroing out all components with $|\boldsymbol{\kappa}| < \kappa_N$, and errors are subsequently computed in the physical domain.

Table 4.1 presents the sub- and super-Nyquist errors for the 2D HIT case reconstructed using CC, WOF, and several NOF variants. At this point, two additional NOF variants are introduced: time-resolved NOF (NOF-TR) and physics-constrained NOF (NOF-phys). NOF-TR extends the single-shot NOF to a spatio-temporally resolved representation (2D+t), while NOF-phys integrates the 2D Navier–Stokes and continuity losses, as defined in Eq. (2.29). Both NOF-TR and NOF-phys were trained on a sequence of 20 particle images, with the reported errors corresponding to the fifth frame, aligning with the non-time-resolved tests.

Among all methods, WOF displays the highest super-Nyquist error, likely due to artifacts introduced by regularization in the wavelet domain. The effects of NOF modifications are also apparent: while NOF-TR incurs slightly higher sub- and super-Nyquist errors, attributed to the increased data and sampling density constraints in the larger spatiotemporal domain, it achieves a 20-fold increase in data compression and reduces the training time per frame from 3–7 minutes to 20–45 seconds. More notably, NOF-phys enhances super-Nyquist accuracy, surpassing the performance of NOF-HD with the hard divergence constraint. While the inclusion of Navier–Stokes residuals has only a minor effect on sub-Nyquist errors, it significantly reduces super-Nyquist errors by alleviating spurious high-frequency artifacts. These findings highlight the importance of physics-based constraints, particularly in minimizing non-physical fluctuations in the super-Nyquist range.

In addition to tests conducted at a fixed seeding density of 0.17 ppp, we assessed the performance of CC, WOF, NOF, and NOF-HD across varying particle densities, with results presented in Fig. 4.3. Our analysis focuses on NOF and NOF-HD, as these two variants consistently demonstrate favorable accuracy (see Table 4.1). The left plot in Fig. 4.3 shows NRMSEs for velocity magnitude fields, averaged over ten consecutive snapshots, across a range of particle densities from 0.01 to 0.4 ppp. Example snapshots at various particle densities are displayed in the bottom row of Fig. 3.1.

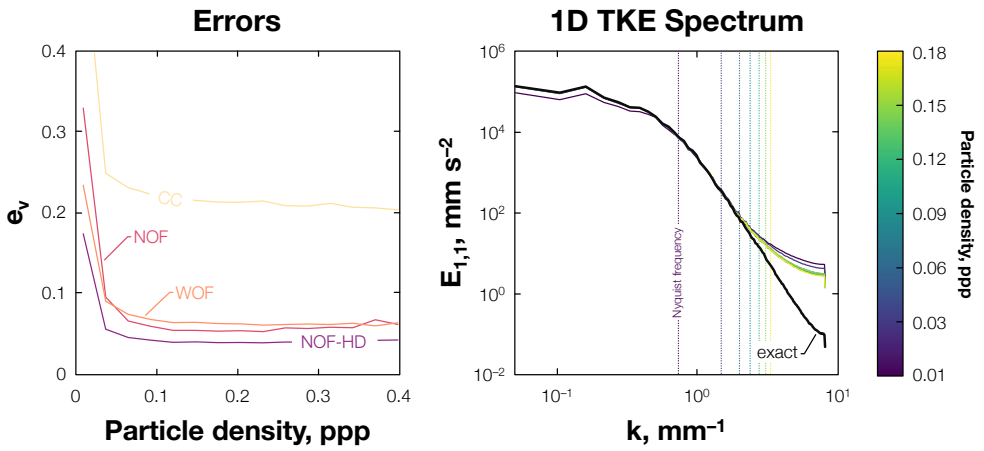

Figure 4.3: NRMSE of 2D HIT flow reconstructed by different methods under varying ppp (left) and the corresponding 1D TKE spectra of NOF reconstruction (right). Note that the NOF's spectra overlap exactly with the ground truth spectrum at ppp > 0.03.

As observed, NOF outperforms WOF at all but the lowest seeding densities and consistently exceeds CC performance across the entire range. NOF-HD delivers the best overall performance, reinforcing the advantage of imposing hard physical constraints for flow reconstruction. The right plot in Fig. 4.3 shows the 1D TKE spectra of NOF-reconstructed fields at different seeding densities. As anticipated, decreasing the density reduces low-wavenumber content while amplifying high-frequency noise. This effect occurs because increased inter-particle spacing allows for more spurious velocity oscillations between particles, which are not penalized by the loss function when the spacing becomes too large.

The plot also includes vertical lines marking the Nyquist frequency for each seeding density, which roughly coincides with the point at which the spectra diverge for ppp values below 0.06. Further increases in seeding density yield minimal improvements in resolving finer flow scales, as the finite size of the particles in the images becomes the limiting factor.

4.2 3D HIT

Both planar and stereo PIV configurations are applied to the 3D HIT flow case. In the planar setup, NOF achieves accuracy comparable to, or slightly better than, WOF. However, in the stereo configuration, NOF significantly outperforms WOF, leveraging the all-in-one flow reconstruction approach described in Sec. 2.3. These tests employ a forward Euler step size of Δt to reduce computational costs, taking advantage of the dataset's small, nearly linear deflections, which permit this simplification.

4.2.1 Planar PIV

Figure 4.4 compares the ground truth DNS velocity magnitude field with reconstructions from CC, WOF, and vanilla NOF. The 3D HIT dataset presents greater challenges than the 2D HIT flow discussed in Sec. 4.1, owing to its high dynamic range and complex spectral content. The intricate flow features within the measurement plane affect the performance of all three methods, yielding

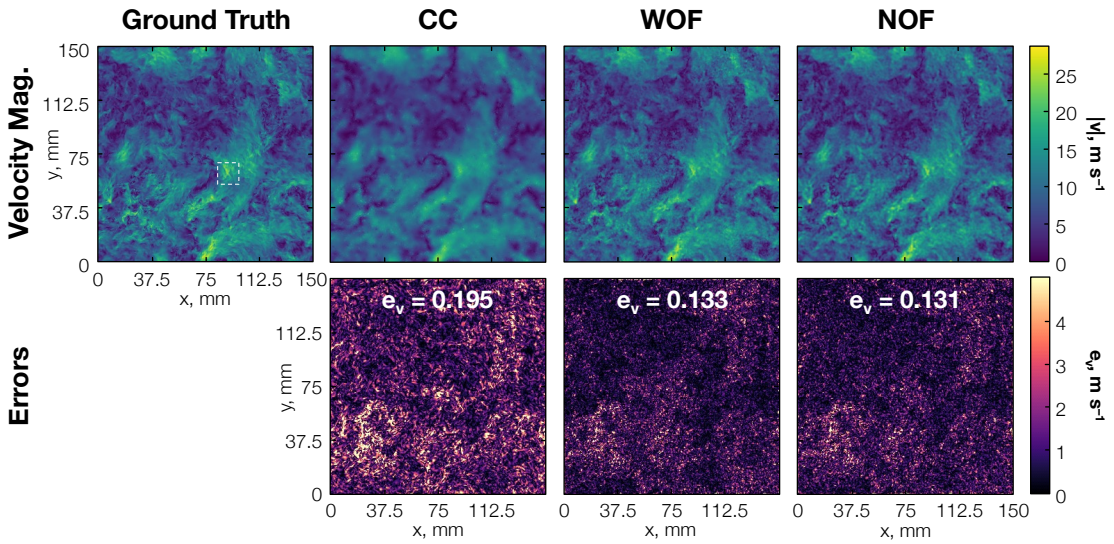

Figure 4.4: Exact and reconstructed 2D2C velocity magnitude fields of 3D HIT with a planar setup (top) and absolute point-wise errors (bottom). NRMSEs are superimposed on the absolute error plot.

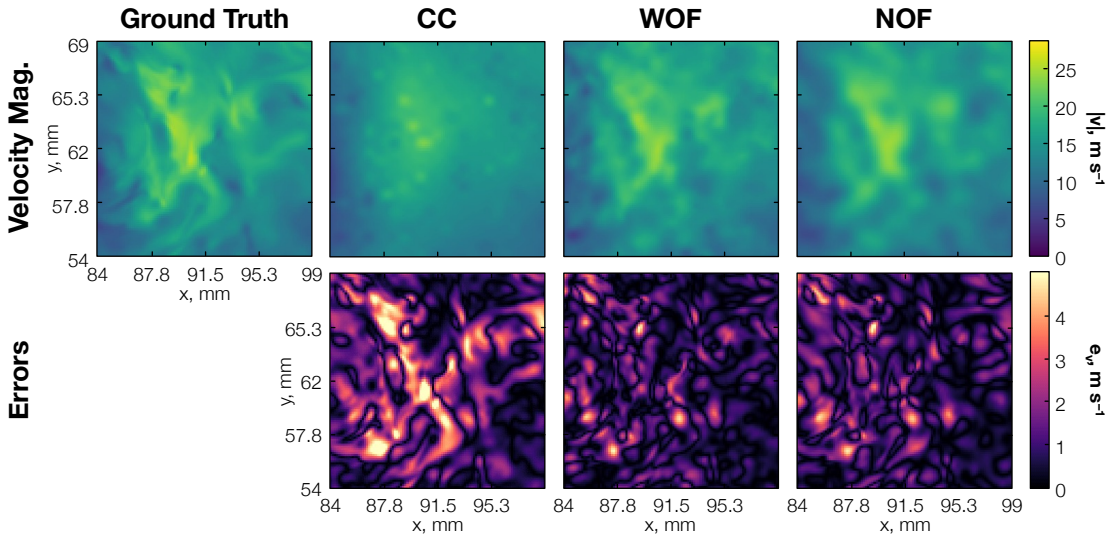

Figure 4.5: Zoomed views of the 2D2C velocity magnitude fields outlined by the white box in Fig. 4.4 (top) and the abolute point-wise errors (bottom).

NRMSE values of 19.5%, 13.3%, and 13.1% for CC, WOF, and NOF, respectively. Although WOF and NOF achieve comparable levels of accuracy, both methods smooth out some of the fine-scale features present in the true field. Notably, WOF *appears* to retain higher-frequency content compared to NOF.

A closer examination of the dashed white square region in Fig. 4.4, as shown in Fig. 4.5, reveals spurious, unresolved features in WOF's reconstruction. In contrast, NOF delivers a smoother and more coherent reconstruction, attributed to the implicit regularization inherent in its neural network design. These zoomed-in views also highlight the limitations of CC: the distinctive "X"-shaped feature present in the ground truth field is entirely absent from the CC estimate. While some

advanced CC methods can resolve finer structures, the implementation used here represents the most commonly applied approach in the field [13, 14, 15, 16, 11, 17, 12, 28, 72].

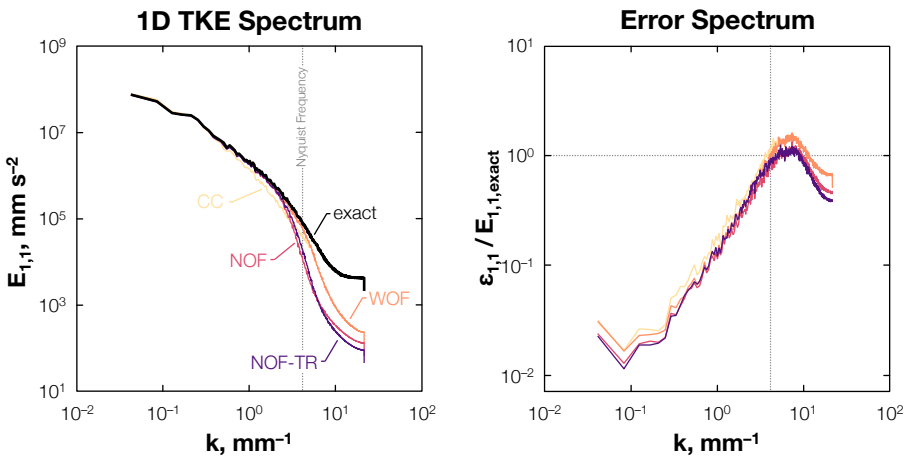

Figure 4.6: Comparison of 1D TKE spectra of reconstructed 2D2C velocity field from a 3D HIT and exact DNS data (left) and normalized error spectra (right).

Figure 4.6 illustrates 1D TKE and normalized error spectra for CC, WOF, NOF, and NOF-TR (evaluated over 20 frames), paralleling the analysis performed for 2D HIT in Fig. 4.2. All methods show spectral deviations from the ground truth before reaching the Nyquist frequency, reflecting their limited capability to fully resolve the flow's high-frequency dynamics. While WOF diverges from the ground truth later than NOF and NOF-TR, this delayed divergence is accompanied by higher errors, as indicated in the right plot. The normalized error spectra reveal that NOF and WOF exhibit similar representational quality across much of the frequency range. However, distinct differences arise at the frequency extremes: at both low and high frequencies, NOF methods outperform the alternatives, demonstrating superior accuracy in these critical regions.

4.2.2 Stereo PIV

Neural optical flow offers a unique integrated approach for stereo flow reconstruction in which a single 3D velocity field is sought to simultaneously minimize both OF data losses. Figure 4.7 compares the out-of-plane velocity fields of the ground truth DNS to CC, WOF, NOF-Recomb. with the standard recombination based stereo methodology, and NOF-Unified with the unified stereo data loss. Unlike CC, WOF, and NOF-Recomb., which combine two separate 2D2C displacement fields into a single 2D3C field using Eq. (A.15), NOF-Unfied directly reconstructs the physical displacement field through a stereo data loss defined in Eq. (2.18). The resulting NRMSEs are 37.5%, 35.9%, 31.4%, and 28.4% for CC, WOF, NOF-Recomb., and NOF-Unified, respectively.

Compared to the planar tests in Sec. 4.2.1, the discrepancies in error between the standard methods and NOF magnify, highlighting their limitations in stereo configurations for capturing out-of-plane motion. Furthermore, the unified method retrieves significantly superior out-of-plane velocity fields, demonstrating its robustness in stereo analysis. The magnified discrepancy in errors is likely due to the recombination step in traditional stereo PIV, which amplifies artifacts in the 2D2C estimates; this is also supported by the difference between NOF-Recomb. and NOF-Unified. As in Sec. 4.2.1, WOF may appear to resolve finer details than NOF, but these details are often

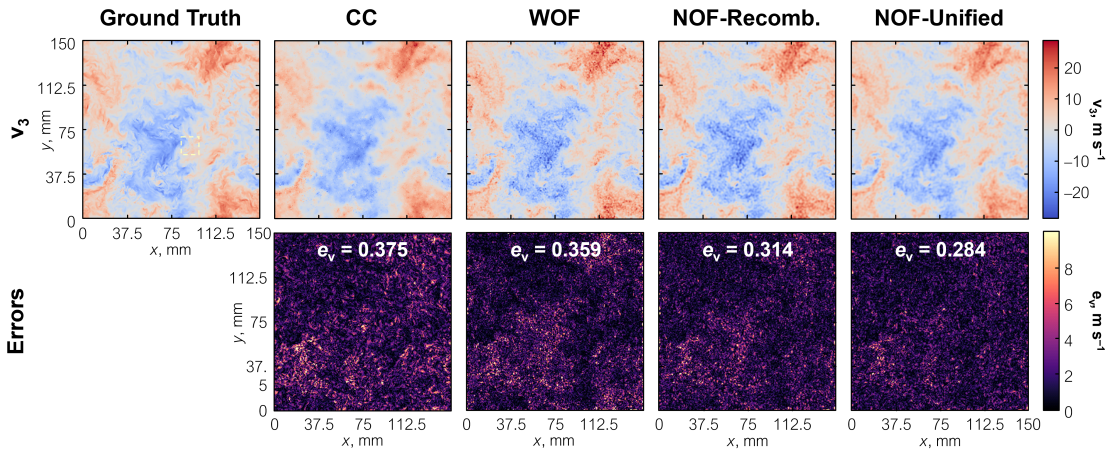

Figure 4.7: Exact and reconstructed 2D3C velocity magnitude fields of 3D HIT with a stereo setup (top) and absolute point-wise errors (bottom). NRMSEs are superimposed on the absolute error plot.

spurious. Figure 4.8 provides close-ups of the fields, highlighting artifacts in the WOF estimate near the left boundary. Comparing these close-ups with Figs. 4.5 and 4.8 further illustrates how the out-of-plane velocity component amplifies these artifacts.

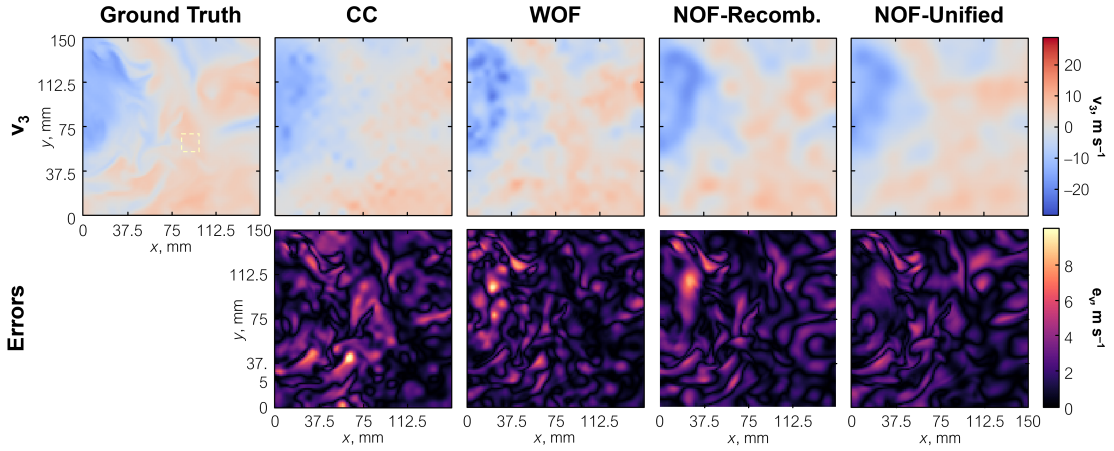

Figure 4.8: Zoomed views of the 2D3C velocity magnitude fields outlined by the white box in Fig. 4.7 (top) and the abolute point-wise errors (bottom).

Figure 4.9 presents a detailed analysis of spectral content through the 1D TKE and normalized error spectra for the v_1 (horizontal) and v_3 (out-of-plane) velocity components. Since the spectral behavior of the two in-plane components, v_1 and v_2 (vertical), is nearly identical, only v_1 is displayed. This figure includes NOF-TR alongside CC, WOF, and vanilla NOF for comparison, with all NOF methods utilizing the unified stereo data loss. The normalized error spectra show a consistent reduction in error across all three velocity components with NOF compared to CC and WOF. For the in-plane components, v_1 and v_2, error reductions are most pronounced at high frequencies, indicating better resolution of fine-scale flow structures. The greatest error reduction, however, occurs in the out-of-plane component, v_3, where NOF achieves lower errors across the entire frequency range. These findings underscore NOF's superior capability in capturing out-of-plane motion, a critical advantage in complex 3D flow scenarios.

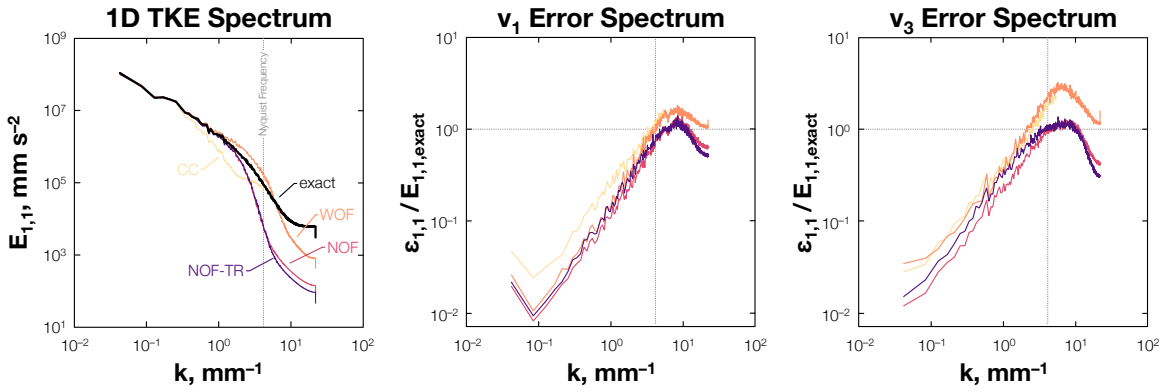

Figure 4.9: Comparison of 1D TKE spectra of reconstructed 2D3C velocity field from a 3D HIT and exact DNS data (left) and normalized error spectra for the v_1 (middle) and v_3 components (right).

A key objective of stereo NOF is to enable pressure recovery from PIV in complex 3D flows. However, in this 3D HIT case, the absence of out-of-plane velocity gradients prevents accurate pressure estimation. Similarly, without $\partial v_3/\partial x_3$ information, the continuity constraint could become inadequate. The physics-informed NOF variants (NOF-HD and NOF-phys) are thus omitted from this analysis. Both NOF and NOF-TR diverge from the ground truth spectrum at the same frequency as WOF, but their spectral content decays more rapidly beyond this point, likely due to limited expressivity of the networks.

4.3 Cylinder Flow

In this section, we present results from the cylinder flow test case, emphasizing pressure reconstruction. NOF's unique capability to *directly* infer pressure from PIV images using an OF-based method distinguishes it from traditional techniques. Figure 4.10 compares the exact and reconstructed vorticity and pressure fields, revealing strong alignment between the reconstructed and ground truth data. Key flow features, such as vortical structures, pressure drops within vortex centers, and the absence of vorticity in regions of mean flow dominance, are accurately captured. Minor discrepancies are observed in regions with steep vorticity gradients, where slight smoothing is evident. Nevertheless, the low NRMSE values (8.28% for vorticity and 8.18% for pressure) confirm the high accuracy of the reconstructions.

Although a pressure Poisson solver is capable of estimating the pressure field from the velocity field in this pseudo-2D flow, this approach often amplifies errors; inaccuracies in the velocity field propagate into the pressure solution without a mechanism to refine the velocity based on the pressure field. Regularization is typically required to address these challenges [73, 74]. By contrast, NOF-phys simultaneously reconstructs pressure and velocity fields, enabling mutual refinement through Navier–Stokes residuals embedded in the penalty term. This integrated approach minimizes inconsistencies, ensuring both fields remain consistent with governing physical laws and the observed image data.

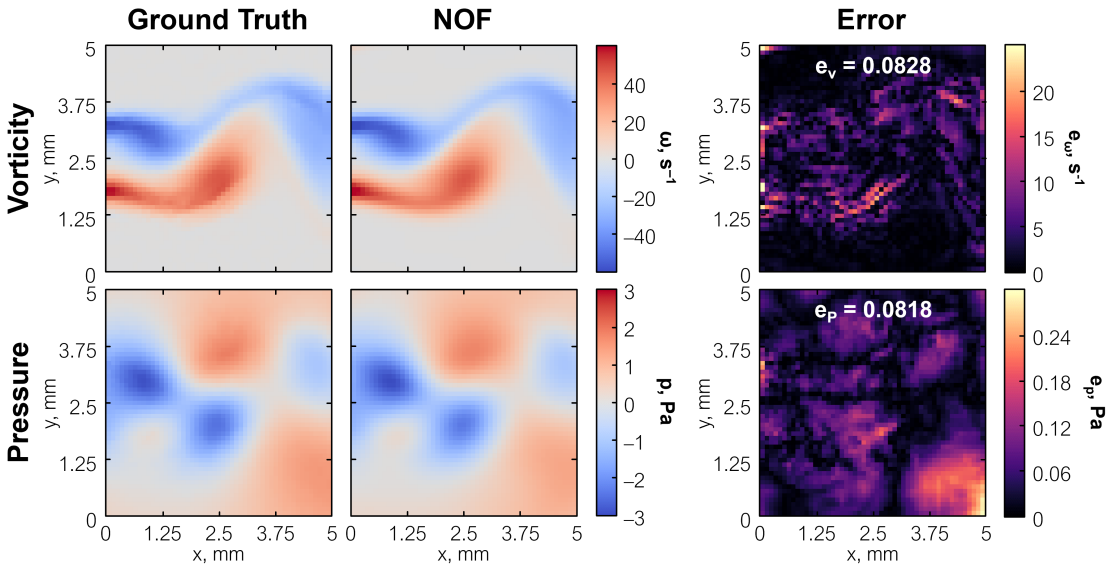

Figure 4.10: Exact and reconstructed cylinder flow vorticity fields (top) and pressure fields (bottom) with corresponding point-wise error fields (right). NRMSEs are superimposed on the absolute error plot.

4.4 Buoyancy Driven Turbulence

In this section, we present results from our BOS buoyancy driven turbulence test case. It was important for us to validate NOF's performance on a task other than PIV and NOF's ability to reconstruct accurate deflection fields from BOS imagery satisfies that need. Figure 4.11 shows a comparison between the exact and reconstructed deflection magnitude fields along with the associated errors. Alignment with the ground truth fields is strong within both WOF and NOF, both qualitatively and quantitatively, capturing most key features, only missing the regions exhibiting high gradients, as seen in the error fields. The NRMSE values of 44.0% for CC, 32.9% for WOF, and 32.0% for NOF affirms the accuracy of NOF and its ability to perform at or slightly above the performance of WOF even in non-PIV deflection sensing scenarios.

An interesting effect to note is that, due to NOF's ability to measure very high displacements accurately and the dual sinusoidal background pattern chosen, NOF sometimes estimates the entire deflection field with a constant offset equal to some multiple of the period of the sinusoid background pattern. This affirms our assertion that NOF handles large displacements very well, but may make the use of such periodic background patterns less suitable for the method, without some penalization of displacement magnitude. An additional hope was that an irrotational hard constraint would improve estimation accuracy, as the deflection fields are irrotaional. This modification does work in some scenarios, and does perform considerably better than the standard methods when used; however, its current implementation is not reliable, so we refrain from showing results.

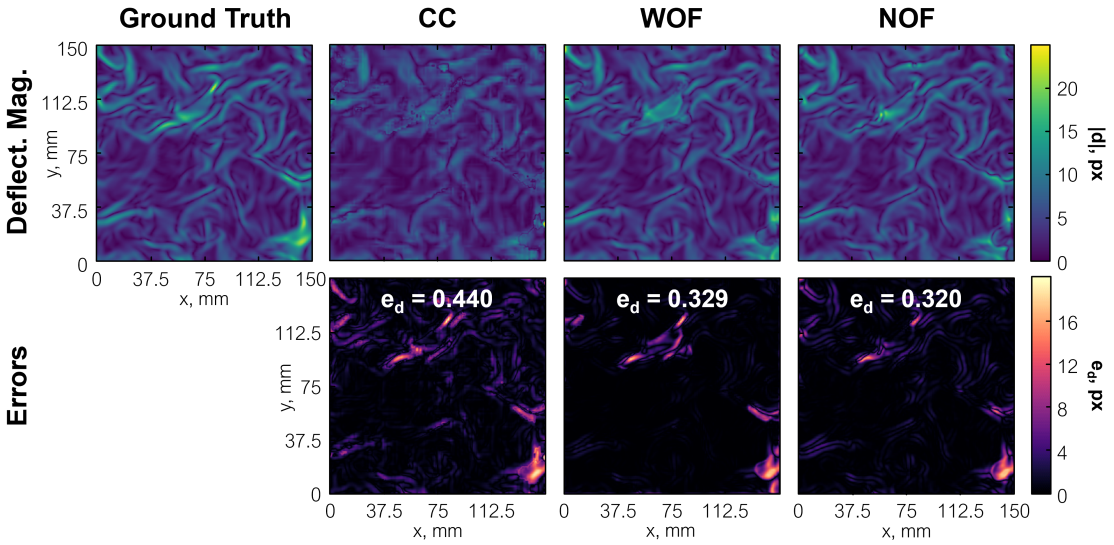

Figure 4.11: Exact and reconstructed bouyancy driven turbulence BOS deflection magnitude fields (top) and absolute point-wise errors (bottom). NRMSEs are superimposed on the absolute error plot.

Chapter 5

Experimental Results

5.1 Vortex Flow

Next, we discuss experimental results for a vortex flow in the wake of an airfoil. Figure 5.1 compares the velocity magnitude and vorticity fields reconstructed by CC, WOF, and NOF. While selecting hyperparameters for CC and NOF was straightforward, WOF demanded extensive tuning due to its sensitivity to noise. Over 30 hyperparameter combinations were tested for WOF before choosing what we considered the best result, though artifacts still appear near the corner of the field. Many other combinations produced larger artifacts closer to the vortex core. Both CC and WOF exhibit grainy textures due to image noise, whereas NOF is more resilient to these effects and produces a smoother field. The vorticity fields highlight this noise sensitivity, particularly near the vortex core, where both CC and WOF produce artifacts that distort the expected vortex shape.

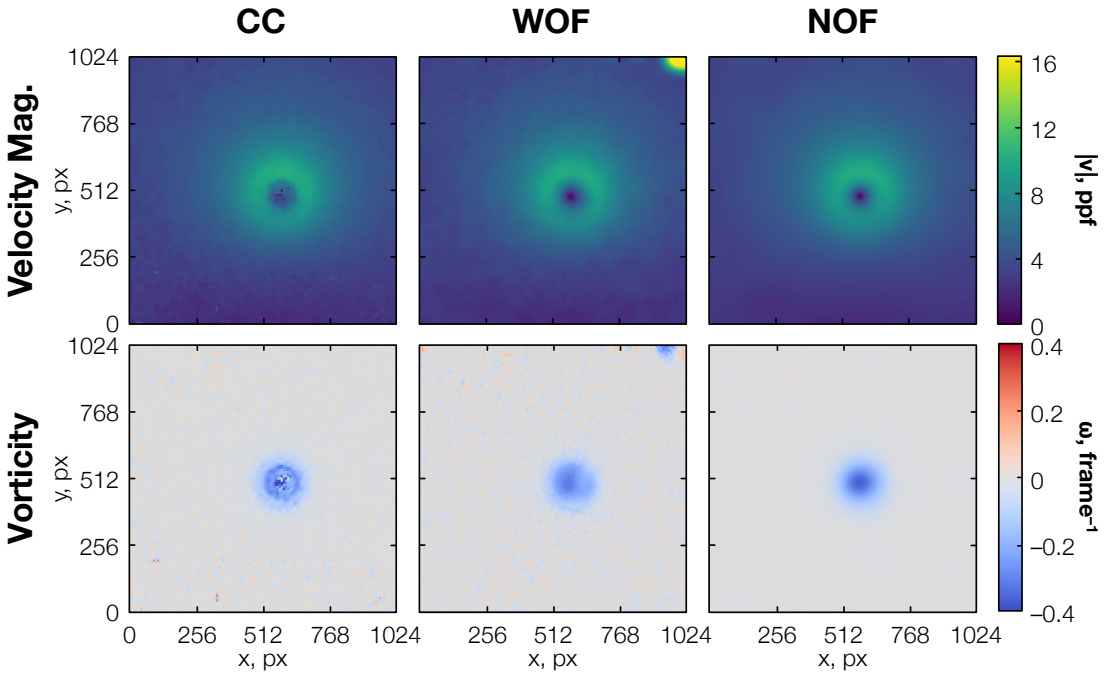

Figure 5.1: Reconstructed vortex flow velocity magnitude fields (top) and vorticity fields (bottom).

Figure 5.2 showcases horizontal cut plots of both velocity components and vorticity, taken slightly offset from the vortex core at $x_2 = 490$ px. The vorticity plot reveals the impact of noise on the reconstructions: while CC and WOF exhibit *a few* similar spikes, most of them are inconsistent between the two methods, suggesting they are in fact non-physical artifacts. Additionally, both the v_1 and vorticity cut plots display clear artifacts within the vortex core, likely due to tracer thinning present in the core of the vortex of these images, implying that the particles are subject to inertial transport rather than passively following the surrounding fluid. Tracer thinning complicates the accurate reconstruction of flow in the core region, and the inertial behavior means the reconstructed fields will be particle velocity fields, rather than fluid velocity fields.

Although it is difficult to decisively determine if NOF provides a more accurate reconstruction due to the absence of ground truth data, NOF aligns more closely with the expected analytical form of a vortex than either CC or WOF. Further research is needed to quantitatively assess the impact of noise on NOF's performance, although these initial results indicate that NOF may be

more resilient to noise than CC or WOF.

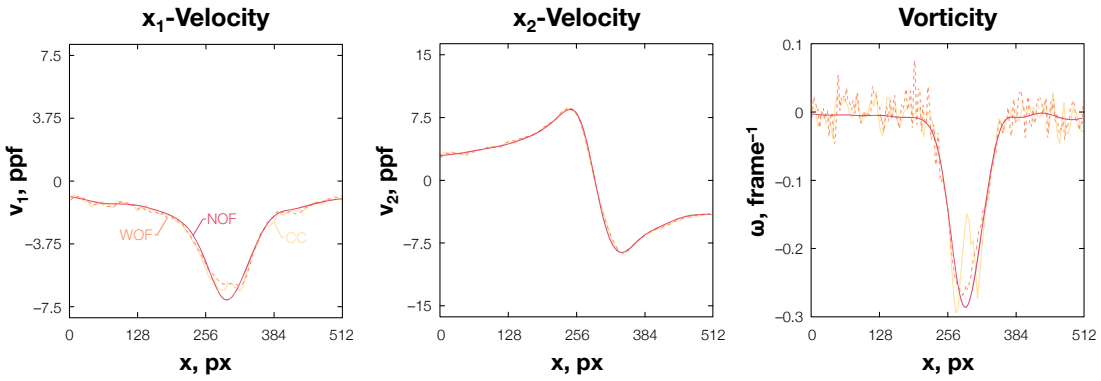

Figure 5.2: Reconstructed vortex flow v_1 cutplots (left), v_2 cutplots (middle), and vorticity cutplots (right).

5.2 Turbulent Jet Flow

We now present the results for the turbulent jet flow case. By having significant energy across a wide range of frequencies, much like the 3D HIT field, hyperparameter selection is more difficult in this scenario. Requiring a large Fourier encoding to capture the flow's dynamics accurately, the added expressivity needed for this representation makes NOF more susceptible to noise-related artifacts. To address this, we applied the div–curl regularization penalty from Eq. (2.22), which mitigates noise artifacts while preserving the network's ability to represent the flow accurately.

Figure 5.3 shows the velocity magnitude and vorticity fields, along with a zoomed-in view of the vorticity field within a region of interest, marked by the dashed box. To enhance visualization, we masked the surrounding flow area outside the jet, where no particles were present (see Fig. 3.1). These masks were created by Gaussian blurring the particle images and applying a threshold determined empirically.

As shown in Fig. 5.3, pronounced "edge effects" are present in the WOF field where the flow transitions from a tracer-rich region to a tracer-free one. When comparing the vorticity fields more broadly, it is revealed that NOF and CC generate more consistent structures, while WOF introduces (what appear to be) spurious fluctuations, similar to those observed in Sec. 4.2. The close-up of the vorticity field highlights an edge effect unique to WOF, absent in both NOF and CC. Given the lack of particle information in this region (see Fig. 3.1), it is reasonable to speculate that NOF and CC are producing more reliable estimates at the periphery of the jet, while WOF's artifacts are likely caused by sensitivity to variable seeding density across the image.

Although it is difficult to conclusively determine which method performs best with experimental data, WOF and NOF likely show similar error patterns, as indicated by the synthetic results in Sec. 4.2, especially given the comparable turbulence and spectral content of these cases. Further research is needed to quantify the impact of noise on NOF's performance, particularly in comparison to state-of-the-art methods like WOF, in order to gain a clearer understanding of their respective strengths and limitations in experimental settings.

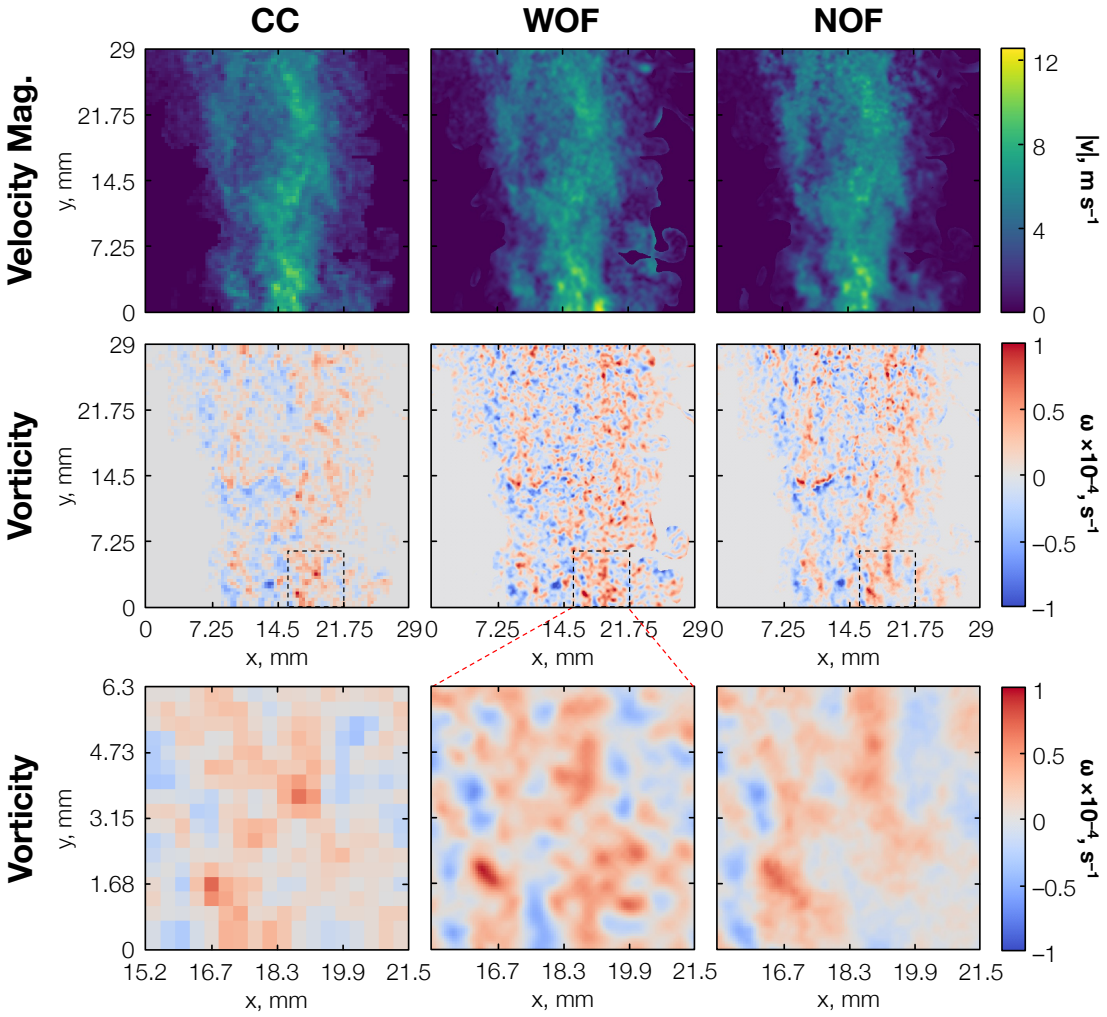

Figure 5.3: Reconstructed jet flow velocity magnitude fields (top), vorticity fields (middle), and vorticity field inserts (bottom).

Chapter 6

Conclusions and Future Work

This manuscript introduces *neural optical flow* (NOF), a neural-implicit method for optical flow (OF) for fluid dynamics research, such as particle image velocimery (PIV) or background oriented schlieren (BOS), aimed at improving velocity reconstruction for turbulent flows while providing a robust framework for data assimilation in PIV. NOF uses a coordinate-based neural network and a differentiable, nonlinear image-warping operator, eliminating the need for a multi-resolution scheme. Data assimilation is facilitated and simplified by the neural implicit basis used, enabling direct pressure reconstruction for 2D and pseudo-2D flows. Furthermore, NOF offers an unified stereo PIV framework that overcomes the limitations of traditional stereo PIV algorithms, which rely on error laden recombinations of separate 2D2C velocity estimates.

We evaluated NOF on both synthetic and experimental datasets, comparing its performance to wavelet-based optical flow (WOF) and cross-correlation (CC), two state-of-the-art PIV techniques. In summary, our key findings are:

1. Vanilla NOF facilitates slightly better velocity reconstructions compared to WOF in planar PIV. This is due to its nonlinear image-warping scheme and implicit suppression spurious high-frequency content.
2. Constraining NOF to solenoidal solutions by applying a hard constraint on mass continuity for incompressible 2D/pseudo-2D flows enhances reconstruction quality, significantly reducing artifacts in high-frequency regions, particularly above the Nyquist frequency.
3. NOF's all-in-one stereo reconstruction framework eliminates the error-amplifying recombination step required in traditional stereo algorithms. Additionally, the out-of-plane energy spectra of the unified stereo NOF are free of spurious high-frequency artifacts, further validating the accuracy of our method.
4. NOF enables the direct inference of latent fields such as pressure, which is demonstrated in Sec. 4.3. Data assimilation faciliated by the neural implicit architecture enables pressure reconstruction from PIV and improves velocity estimates, as quantified in Table 4.1, offering a robust alternative to post-processing with a pressure Poisson solver.

In summary, NOF offers a versatile and adaptable OF framework for PIV and addresses a broad range of fluid measurement challenges, including stereo PIV, pressure from PIV, and deflection sensing within BOS. While its performance matches or slightly surpasses existing state-of-the-art methods in most cases, NOF provides additional benefits including more accurate and robust stereo, smoother reconstructions in situations with limited, high-frequency data, enhanced feature extraction, and more accurate derivative-based flow statistics, especially in turbulent flows.

Appendix A

Camera Transform

A.1 Forward Transform

Projecting points from world space to image space requires a camera model. We use the pinhole camera which often provides sufficient accuracy for PIV imaging. For a distortion-free pinhole model, the forward camera transform is defined as

$$\mathbf{s} = \zeta^{-1}\mathbf{K}\left(\mathbf{R}\mathbf{x} + \mathbf{t}\right). \tag{A.1}$$

where $\mathbf{K}$ is the 2×3 intrinsic matrix, $\mathbf{R}$ is a 3×3 orthonormal rotation matrix, and $\mathbf{t}$ is the translation vector. This vector is defined as $\mathbf{t} = -\mathbf{R}\mathbf{c}$, where $\mathbf{c}$ is the camera's position in world coordinates. The scale factor ζ ensures projection onto the image plane and is defined as

$$\zeta = \mathbf{R}_{3,*}\mathbf{x} + t_3, \tag{A.2}$$

where $\mathbf{R}_{3,*}$ is the third row of $\mathbf{R}$ and t_3 is the third element of $\mathbf{t}$.

The intrinsic parameters in $\mathbf{K}$ include the distance from the pinhole to the sensor, $\hat{f}$, which defines the focal length of a pinhole camera, and the principal point, $\mathbf{s}_0 = (s_{0,1}, s_{0,2})$, where the optical axis intersects the sensor. Using these elements, the intrinsic matrix is

$$\mathbf{K} = \begin{bmatrix} \psi_1\hat{f} & 0 & s_{0,1} \\ 0 & \psi_2\hat{f} & s_{0,2} \end{bmatrix}. \tag{A.3}$$

The terms $\psi_1\hat{f}$ and $\psi_2\hat{f}$ represent the effective focal lengths in pixel units along the horizontal and vertical directions; for square pixels, $\psi_1 = \psi_2 = \psi$. Note that the focal length of a pinhole camera, $\hat{f}$, differs from that of a physical lens, f. The relationship between these quantities is given by

$$\hat{f} = f\left(1 + \frac{f}{d_\mathrm{o} - f}\right), \tag{A.4}$$

where d_o is the distance from the aperture (or pinhole) to the object plane [75].

Lens distortions can be incorporated using a distortion model, typically applied to the normalized image coordinates, i.e., before scaling by the intrinsic matrix,

$$\mathbf{s}' = \zeta^{-1}\left(\mathbf{R}\mathbf{x} + \mathbf{t}\right), \tag{A.5}$$

where $(\cdot)'$ indicates a normalized quantity and $\mathbf{s} = \mathbf{K}\mathbf{s}'$. Radial and tangential distortions are computed using Brown's polynomial distortion model [76, 77],

$$\mathbf{s}'_\mathrm{dist} = \mathcal{D}(\mathbf{s}'). \tag{A.6}$$

This function includes additional intrinsic parameters: specifically, a set of radial and tangential distortion coefficients. Bringing these elements together, the transform for a pinhole camera with distortions is

$$\mathbf{s} = \underbrace{\mathbf{K}\,\mathcal{D}\Big[\zeta^{-1}(\mathbf{R}\mathrm{x} + \mathbf{t})\Big]}_{\Psi(\mathbf{x})}. \tag{A.7}$$

In practice, the intrinsic and extrinsic parameters in $\mathbf{K}$, $\mathbf{R}$, $\mathbf{t}$, and $\mathcal{D}$ must be obtained through a camera calibration procedure, while ψ is a fixed physical property of the sensor.

A.2 Inverse Transform

The inverse mapping from sensor coordinates to world coordinates, $\mathbf{s}$ to $\mathbf{x}$, involves inverting the forward projection function in Eq. (A.7). The forward transformation is not generally invertible, as each pixel corresponds to a line through world space, rather than a single point. By specifying the distance along this line, l, a unique 3D position can be found. In planar and stereo PIV, this is the distance where the line intersects the laser sheet, as discussed in the following sections. In volumetric PIV, some trivial modifications removing the inverse mapping are required, as inversion would be impossible.

The general inverse transform is

$$\mathbf{x} = \underbrace{\frac{l}{\|\mathbf{r}\|_2} \overbrace{\mathbf{R}^\top \begin{bmatrix} \mathbf{K}^{-1}\mathbf{s} - \mathbf{s}_0' \\ 1 \end{bmatrix}}^{\mathbf{r}} + \mathbf{c}}_{\Psi^{-1}(\mathbf{s},l)}, \tag{A.8}$$

where $\mathbf{r}$ is a ray-direction vector; $\mathbf{K}^{-1}$ and $\mathbf{s}_0'$ are defined as

$$\mathbf{K}^{-1} \equiv \hat{f}^{-1} \begin{bmatrix} \psi_1^{-1} & 0 \\ 0 & \psi_2^{-1} \end{bmatrix} \quad \text{and} \quad \mathbf{s}_0' \equiv \hat{f}^{-1} \begin{bmatrix} s_{0,1}/\psi_1 \\ s_{0,2}/\psi_2 \end{bmatrix}. \tag{A.9}$$

When lens distortions are significant, the undistorted sensor coordinates must be found by inverting the distortion model,

$$\mathbf{s}_{\text{dist}} = \mathbf{K}\,\mathcal{D}\Big(\mathbf{K}^{-1}\mathbf{s} - \mathbf{s}_0'\Big), \tag{A.10}$$

which is iteratively solved. Here, $\mathbf{s}_{\text{dist}} = \mathbf{K}\mathbf{s}'_{\text{dist}}$ is a pixel centroid and the solution $\mathbf{s}$ is used in Eq. (A.8).

A.3 Application to Planar PIV

In planar PIV, the camera is positioned perpendicular to the laser sheet, with the optical axis parallel to the x_3-axis of the world coordinate system. The world origin is set at the center of the laser sheet, and the camera's position is $\mathbf{c} = (0, 0, -d_\text{o})$, where d_o is the object distance. Since the world and camera coordinate systems are perfectly aligned, $\mathbf{R}$ is the identity matrix and the problem can be stated in a simpler manner.

Assuming no distortion and that the measurement domain lies within the $x_1{-}x_2$ plane (i.e., $x_3 = 0$), the forward camera transform simplifies to

$$\mathbf{s} = \left(\psi_1 \hat{f}\frac{x_1}{d_\text{o}} + s_{0,1},\ \psi_2 \hat{f}\frac{x_2}{d_\text{o}} + s_{0,2}\right) \tag{A.11a}$$

and the inverse transform is given by

$$\mathbf{x} = \frac{d_\text{o}}{\hat{f}}\left(\frac{s_1 - s_{0,1}}{\psi_1},\ \frac{s_2 - s_{0,2}}{\psi_2},\ 0\right). \tag{A.11b}$$

While minor misalignments and lens distortions are inevitable in a real experiment, they can often be neglected in planar PIV. The full transforms in A.1 and A.2 can be used to handle cases with significant alignment and distortion issues.

A.4 Application to Stereo PIV

A.4.1 Translational Configuration

The translational stereo PIV configuration builds on conventional planar PIV by using two cameras positioned to capture the laser sheet from slightly different perspectives. The object plane, lens plane, and sensor planes of both cameras are all parallel. In a symmetric configuration (see Fig. 2.2), the left and right camera positions in world space are

$$\mathbf{c}_{\mathrm{L}} = \left(-\frac{L}{2},\ 0,\ -d_{\mathrm{o}}\right) \tag{A.12a}$$

and

$$\mathbf{c}_{\mathrm{R}} = \left(\frac{L}{2},\ 0,\ -d_{\mathrm{o}}\right), \tag{A.12b}$$

where L is the distance between the camera lenses, and subscripts "L" and "R" indicate the left and right cameras. Assuming perfect alignment with no lens distortion, $\mathbf{R} = \mathbf{I}$ and the forward camera transforms become

$$\mathbf{s}_{\mathrm{L}} = \left(\psi_1 \hat{f} \frac{x_1 + L/2}{x_3 + d_{\mathrm{o}}} + s_{0,1},\ \psi_2 \hat{f} \frac{x_2}{x_3 + d_{\mathrm{o}}} + s_{0,2}\right) \tag{A.13a}$$

and

$$\mathbf{s}_{\mathrm{R}} = \left(\psi_1 \hat{f} \frac{x_1 - L/2}{x_3 + d_{\mathrm{o}}} + s_{0,1},\ \psi_2 \hat{f} \frac{x_2}{x_3 + d_{\mathrm{o}}} + s_{0,2}\right) \tag{A.13b}$$

These equations are consistent with previous derivations [78, 4]. The inverse transforms, which map image coordinates back to world coordinates, are given by

$$\mathbf{x}_{\mathrm{L}} = \left[(d_{\mathrm{o}} + x_3)\,\frac{s_1 - s_{0,1}}{\psi_1 \hat{f}} - \frac{L}{2},\ (d_{\mathrm{o}} + x_3)\,\frac{s_2 - s_{0,2}}{\psi_2 \hat{f}},\ x_3\right] \tag{A.14a}$$

and

$$\mathbf{x}_{\mathrm{R}} = \left[(d_{\mathrm{o}} + x_3)\,\frac{s_1 - s_{0,1}}{\psi_1 \hat{f}} + \frac{L}{2},\ (d_{\mathrm{o}} + x_3)\,\frac{s_2 - s_{0,2}}{\psi_2 \hat{f}},\ x_3\right]. \tag{A.14b}$$

While particles may be advected out of the measurement plane ($x_3 \neq 0$), stereo PIV assumes that the particles remain within the laser sheet, so we restrict all initial particle positions to the x_1–x_2 plane, with $x_3 = 0$. This simplification aligns with the planar assumption commonly used in stereo PIV.

Conventional stereo PIV combines two 2D2C displacement fields, computed from each camera view using a method like CC or OF, into a single 2D3C displacement field in world space. Assuming square pixels, $\psi_1 = \psi_2 = \psi$, the relationship between the physical vector, $\Delta\mathbf{x} = (\Delta x_1, \Delta x_2, \Delta x_3)$,

and the image displacement vectors, $\Delta\mathbf{s}_\mathrm{L}$ and $\Delta\mathbf{s}_\mathrm{R}$, can be expressed as

$$\Delta x_1 = \frac{\Delta s_{\mathrm{R},1}\,(x_1 + L/2) - \Delta s_{\mathrm{L},1}\,(x_1 - L/2)}{\psi ML + (\Delta s_{\mathrm{L},1} - \Delta s_{\mathrm{R},1})} \tag{A.15a}$$

$$\Delta x_2 = \frac{x_2 \Delta x_3}{d_\mathrm{o}} + \frac{\Delta s_{\mathrm{L},2} + \Delta s_{\mathrm{R},2}}{2\psi M}\left(\frac{\Delta x_3}{d_\mathrm{o}} + 1\right) \tag{A.15b}$$

$$\Delta x_3 = \frac{-d_\mathrm{o}(\Delta s_{\mathrm{L},1} - \Delta s_{\mathrm{R},1})}{\psi ML + (\Delta s_{\mathrm{L},1} - \Delta s_{\mathrm{R},1})}. \tag{A.15c}$$

Since each 3C displacement vector in world space corresponds to two 2C displacement vectors in image space (four pieces of information), the stereo analysis is overdetermined. This redundancy can be leveraged to improve the accuracy of Eq. (A.15b), e.g., by averaging $\Delta s_{\mathrm{L},2}$ and $\Delta s_{\mathrm{R},2}$, explaining the asymmetry between Eqs. (A.15a) and (A.15b).

A.4.2 Rotational Configuration

Rotational stereo PIV employs inward-facing cameras, each aimed at the center of the laser sheet. Scheimpflug adapters are commonly used to ensure that the entire sheet remains in focus.[1] When the Scheimpflug condition is met, the object, lens, and sensor planes all intersect along a common line. This contrasts with the standard pinhole model in Eq. (A.1), which assumes parallel alignment between the lens and sensor planes.

Cornic et al. [81] developed a modified pinhole model to account for the sensor tilt introduced by a Scheimpflug adapter,

$$\mathbf{s} = \zeta^{-1}\mathbf{K}\mathbf{S}\mathbf{R}_\alpha\,(\mathbf{R}\mathbf{x} + \mathbf{t})\,, \tag{A.16}$$

where $\mathbf{R}_\alpha$ is the rotation matrix accounting for sensor tilt, $\mathbf{S}$ is the Scheimpflug matrix, and

$$\zeta = (0, 0, 1) \cdot [\mathbf{S}\mathbf{R}_\alpha\,(\mathbf{R}\mathbf{x} + \mathbf{t})] \tag{A.17}$$

is the scale factor that projects world coordinates onto the image plane. The sensor rotation matrix $\mathbf{R}_\alpha$ can be decomposed into rotations about the s_1 and s_2 axes, with respective tilt angles α_1 and α_2,

$$\mathbf{R}_\alpha = \begin{bmatrix} \cos(\alpha_2) & 0 & \sin(\alpha_2) \\ 0 & 1 & 0 \\ -\sin(\alpha_2) & 0 & \cos(\alpha_2) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\alpha_1) & \sin(\alpha_1) \\ 0 & -\sin(\alpha_1) & \cos(\alpha_1) \end{bmatrix}. \tag{A.18}$$

The Scheimpflug matrix $\mathbf{S}$ depends on these same tilt angles,

$$\mathbf{S} = \begin{bmatrix} \cos(\alpha_1)\cos(\alpha_2) & 0 & \sin(\alpha_2) \\ 0 & \cos(\alpha_1)\cos(\alpha_2) & -\cos(\alpha_2)\sin(\alpha_1) \\ 0 & 0 & 1 \end{bmatrix}. \tag{A.19}$$

Lens distortions are modeled similarly to Eq. (A.6). Incorporating the distortion model, the forward camera transform becomes

$$\mathbf{s} = \mathbf{K}\,\mathcal{D}\Big[\zeta^{-1}\mathbf{S}\mathbf{R}_\alpha\,(\mathbf{R}\mathbf{x} + \mathbf{t})\Big]. \tag{A.20}$$

[1] Rotational configurations can be implemented without Scheimpflug adapters, though the aperture must be stopped down as far as possible to achieve a sufficient depth-of-focus [79, 59, 80].

For the inverse transform, where the goal is to map from sensor coordinates to world coordinates, we assume a laser sheet at the origin with a normal vector $(0, 0, 1)$. The resultant inverse mapping is

$$\mathbf{x} = -\frac{c_3}{r_3}\mathbf{r} + \mathbf{c}, \tag{A.21a}$$

where the ray direction is

$$\mathbf{r} = \mathbf{R}^\top \mathbf{R}_\alpha^\top \mathbf{S}^{-1} \begin{bmatrix} \mathbf{K}^{-1}\mathbf{s} - \mathbf{s}_0' \\ 1 \end{bmatrix}, \tag{A.21b}$$

Here, c_3 and r_3 are the third components of $\mathbf{c}$ and $\mathbf{r}$, respectively. As before, if lens distortions are present, the distortion model must be inverted to recover the undistorted sensor coordinates using Eq. (A.10).

Appendix B

Network Architecture

In this work, we use a coordinate neural network to represent flow fields in the physical domain $\mathcal{P}$. The network consists of an input layer, an output layer, and a series of n_l hidden layers,

$$\mathbf{z}_\mathrm{out} = \overbrace{\mathbf{W}^{n_\mathrm{l}+1}\Big[\mathcal{L}^{n_\mathrm{l}} \circ \mathcal{L}^{n_\mathrm{l}-1} \circ \cdots \circ \mathcal{L}^{2} \circ \mathcal{F}(\mathbf{z}_\mathrm{in})\Big] + \mathbf{b}^{n_\mathrm{l}+1}}^{\mathcal{N}(\mathbf{z}_\mathrm{in})}, \tag{B.1a}$$

with

$$\mathbf{z}^l = \underbrace{\sigma\Big(\mathbf{W}^l\mathbf{z}^{l-1} + \mathbf{b}^l\Big)}_{\mathcal{L}^l(\mathbf{z}^{l-1})} \quad \text{for} \quad l \in \{2, 3, \ldots, n_\mathrm{l}\}. \tag{B.1b}$$

In this paper, $\mathbf{z}_\mathrm{in}$ is $(\mathbf{x}, t)$ and $\mathbf{z}_\mathrm{out}$ is $\mathbf{v}$.[1] The vector $\mathbf{z}^l$ contains the activation values of neurons in the lth layer, $\mathbf{W}^l$ and $\mathbf{b}^l$ are the weight matrix and bias vector, and σ is a nonlinear activation function that is applied element-wise. We apply the weight normalization technique of Salimans and Kingma [82] to accelerate training; we also use GELU activation functions, which showed comparable performance to other non-saturating functions, such as swish, and outperformed hyperbolic tangent functions and ReLU in preliminary tests.

To mitigate the low-frequency spectral bias inherent in gradient-based optimizers and neural networks especially [83], we replace the first hidden layer, $\mathcal{L}^1$, with a Fourier encoding layer [84],

$$\mathbf{z}^1 = \underbrace{\Big[\sin\Big(2\pi\boldsymbol{\omega}_1 \cdot \mathbf{z}^0\Big),\, \cos\Big(2\pi\boldsymbol{\omega}_1 \cdot \mathbf{z}^0\Big), \ldots,\, \sin\Big(2\pi\boldsymbol{\omega}_w \cdot \mathbf{z}^0\Big),\, \cos\Big(2\pi\boldsymbol{\omega}_w \cdot \mathbf{z}^0\Big)\Big]}_{\mathcal{F}(\mathbf{z}_\mathrm{in})} \tag{B.2}$$

Here, w is the number of Fourier features, and each $\boldsymbol{\omega}$ is a vector of random frequencies, sampled from a normal distribution and fixed before training. This method embeds the input coordinates into a higher dimensional space defined with useful spectral qualities that the neural network can better harness to represent fields with high frequency content. The standard deviation of the frequency distribution controls the spectral representational capacity and, as discussed in Sec. 2.1, acts as an implicit regularization mechanism by introducing a controlled spectral bias, which smooths the fields. For time-resolved data, the spatial and temporal standard deviations are adjusted independently to account for variations in spectral content between spatial and temporal features.

The network size (depth and width) is crucial for ensuring its ability to represent the target function. We tested architectures with 7–12 hidden layers and 200–600 neurons per layer, observing consistent results across all configurations. A network with nine hidden layers and 500 neurons per layer provided sufficient representation across all tested flow cases, and therefore was used throughout the study.

The Adam optimizer [85] was employed with a constant learning rate of 10^{-3}. Training durations varied depending on the flow scenario and are reported throughout the paper on a per-case basis. All experiments were implemented using PyTorch 2.4 on an NVIDIA RTX 3090 GPU.

[1] The output $\mathbf{z}_\mathrm{out}$ is φ or $\boldsymbol{\varphi}$ for 2D and 3D velocity fields with a hard constraint on continuity, as discussed in Sec. 2.5, and pressure is outputted when using a Navier–Stokes penalty term, $\mathbf{z}_\mathrm{out} = (\mathbf{v}, p)$ or (φ, p).

Appendix C

Interpolation Schemes

Approximating the integral in Eq. (2.8a) requires continuous sampling of the intensity field, which is recorded as a discrete image signal. We tested both bilinear and bicubic interpolation schemes, which provide smooth, differentiable representations of the intensity field. These schemes are essential for our image warping operator to perform accurately during optimization.

Let $\mathbf{s}_{i,j}$ be the centroid of the (i, j)th pixel and $I_{i,j}$ be its intensity. The image vector for an $m \times n$ image is $\mathbf{I} = \{I_{i,j} \mid i = 1, \ldots, m, j = 1, \ldots, n\}$. Assuming pixels with a side length of one, the bilinear interpolation operator $\mathcal{B}$ is

$$\hat{I}(\mathbf{s}) = \overbrace{w_1 I_{i,j} + w_2 I_{i+1,j} + w_3 I_{i,j+1} + w_4 I_{i+1,j+1}}^{\mathcal{B}(\mathbf{I},\mathbf{s})}, \tag{C.1a}$$

where the interpolation weights are

$$w_1 = \Big(1 - s_1 + s_{i,j,1}\Big)\Big(1 - s_2 + s_{i,j,2}\Big), \tag{C.1b}$$

$$w_2 = \Big(s_1 - s_{i,j,1}\Big)\Big(1 - s_2 + s_{i,j,2}\Big), \tag{C.1c}$$

$$w_3 = \Big(1 - s_1 + s_{i,j,1}\Big)\Big(s_2 - s_{i,j,2}\Big), \quad \text{and} \tag{C.1d}$$

$$w_4 = \Big(s_1 - s_{i,j,1}\Big)\Big(s_2 - s_{i,j,2}\Big), \tag{C.1e}$$

where $\mathbf{s}$ is the query point, enclosed by the centroids $\mathbf{s}_{i,j}$, $\mathbf{s}_{i+1,j}$, $\mathbf{s}_{i,j+1}$, and $\mathbf{s}_{i+1,j+1}$. This method provides a suitable approximation of intensity across the image.

Bicubic interpolation, on the other hand, uses a 16-pixel neighborhood and computes weights based on polynomial functions of the distances between the query point and pixel centroids. This method captures more detail and provides a higher-order representation than bilinear interpolation, especially near edges of the image. We use the convolution-based implementation from Keys [86], setting the interpolation parameter to -0.75, as recommended for optimal smoothness. Further technical details on both interpolation schemes can be found in *Numerical Recipes* [87].

In our implementation, we use the `grid_sample` function in PyTorch for both forms of interpolation. Bilinear and bicubic interpolation provided indistinguishable results during preliminary testing on particle images; due to its lower computational cost, bilinear interpolation is thus used as a default in this work, unless otherwise stated.

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